

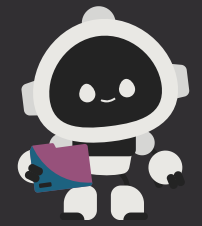
MULTI-MODAL CAPACITATED VEHICLE ROUTING PROBLEM



Group 9
Team members:

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PROBLEM DEFINITION

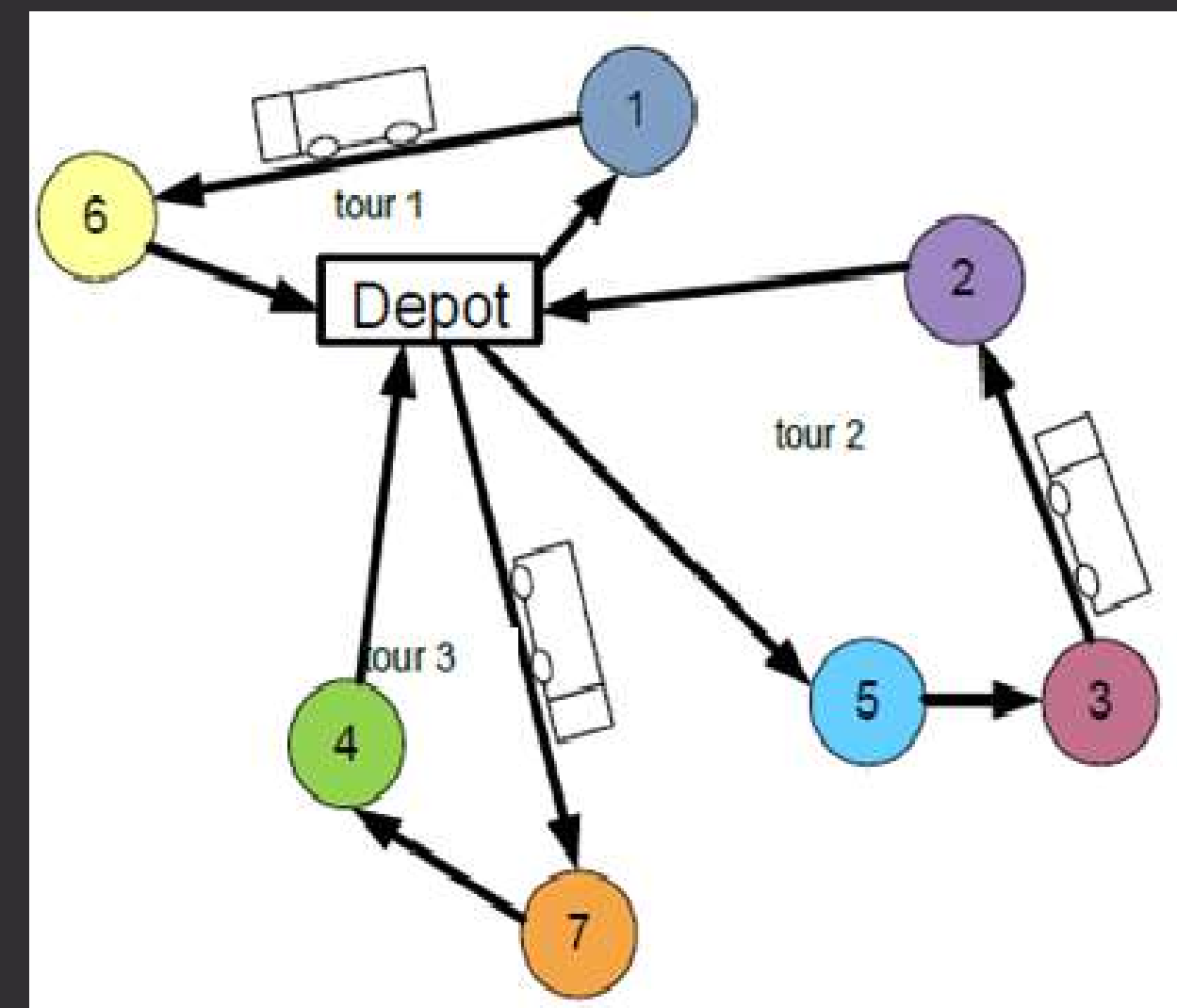


The CVRP consists of designing optimal routes for a fleet of vehicles serving multiple customers from a central depot.

The main objective is to minimize the total distance traveled, ensuring all deliveries are made without exceeding the maximum load capacity of each truck.

A valid solution to the problem must strictly satisfy the following operational constraints:

- Origin and Destination
- Single Visit
- Flow Conservation
- Capacity Constraint

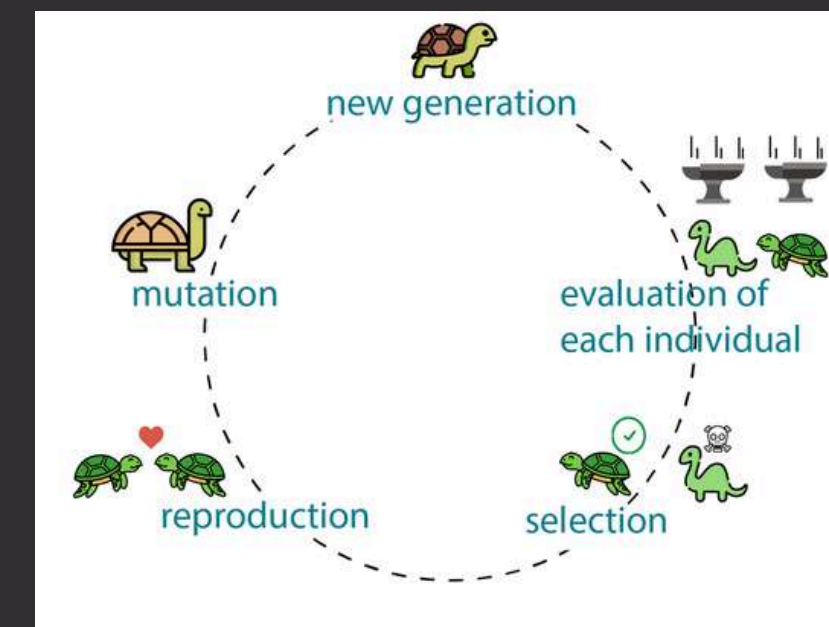
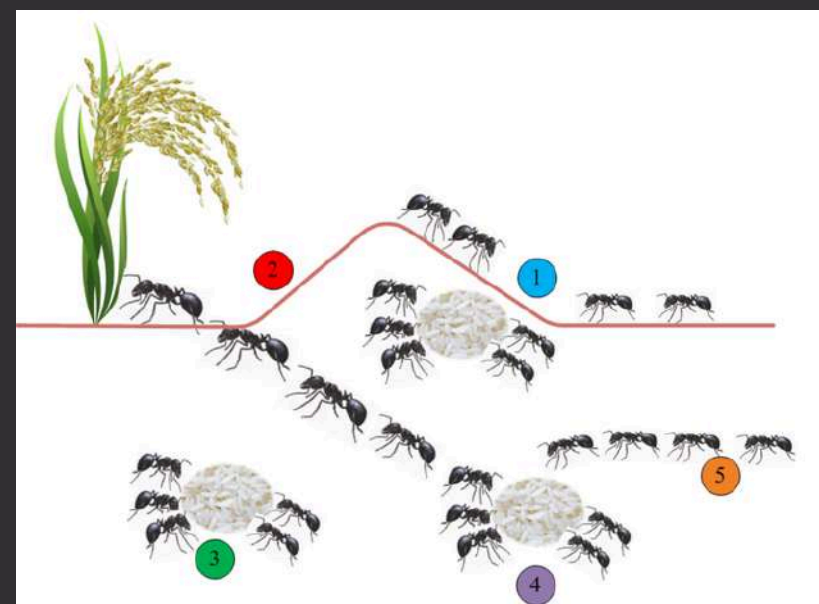


VARIATION OF THE PROBLEM

To handle real-world uncertainties, we applied PSO, GA, and ACO to identify three robust and distinct routing alternatives.

Success Criteria:

- High Quality: All three alternatives must maintain a total distance within a 5% gap of the best-found solution.
- Structural Diversity: Routes must differ significantly in their customer-to-vehicle assignments to provide genuine decision-making options and operational flexibility.



DATASETS

The data required for each problem is:

- Number of delivery nodes (n)
- Spatial coordinates for each node (x, y)
- Specific customer demand per node (d_i)
- Fixed maximum vehicle capacity (C)

Dataset & Setup

- Benchmark: 100 instances from the Uchoa 'X' dataset.
- Complexity: 101 to 916 nodes.
- Evaluations: 5,000 (<200 nodes) or 10,000 (≥ 200 nodes).

Main Case Study (X-n106-k14)

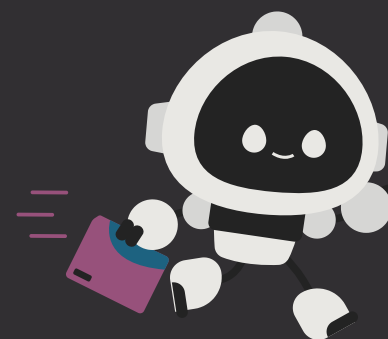
- Optimum Fitness: 26,362
- 5% Limit (Archive): 27,680



CODIFICATION

To tackle the logistical complexity of the problem, we propose a dual-codification architecture tailored to the nature of each metaheuristic, supported by a shared memory system:

- Dual Architecture: We propose using a Continuous Permutation representation for the evolutionary/swarm algorithms (PSO and GA), and a Graph-Based representation for the constructive algorithm (ACO).
- The Multimodal Manager (The Archive): To meet the main challenge, we propose an algorithm-independent "Elite Archive". This storage dynamically verifies and saves solutions, keeping only the Top 3 that maintain a quality gap $< 5\%$ and high structural diversity (Jaccard metric).





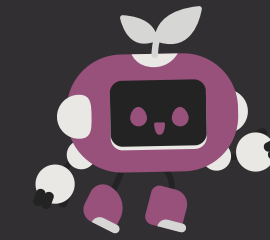
Approach A: Customer Permutation (PSO & GA)

- The Representation: The algorithm generates a continuous list of customers (a "Giant Tour"), initially ignoring the existence of multiple vehicles and the central depot.
 - Code Example: [3, 1, 4, 5, 2]
- The Mechanics (The Split): We propose using a "decoder" that reads the array from left to right. Whenever adding the next customer exceeds the truck's maximum load (C), the code "cuts" the list and inserts a return to the depot (Node 0).
 - Generated Routes: $0 \rightarrow 3 \rightarrow 1 \rightarrow 0$ and $0 \rightarrow 4 \rightarrow 5 \rightarrow 2 \rightarrow 0$

Approach B: Graph-Based Construction (ACO)

- The Representation: Ants build the sequence step-by-step directly on the logistical map, deciding in real-time when to include the depot (Node 0) to separate the routes.
 - Code Example: [0, 3, 1, 0, 4, 5, 2, 0]
- The Mechanics (Hard Constraints): We propose strict feasibility filtering. If an ant tries to visit a customer but its truck lacks capacity, the algorithm physically blocks that path. The ant is forced to select Node 0 to unload before continuing.

FITNESS FUNCTIONS



1. The Core Objective (Routing Cost)

- The Metric: The baseline fitness function for all algorithms is the total distance traveled by the fleet.
- The Goal: Minimizing this value is the fundamental objective

2. Soft Constraints & Penalties

Permits temporary invalid routes to enhance exploration, controlled via massive fitness penalties:

- PSO (Fleet Penalty): Quadratic cost for using excess vehicles.
 $\text{Excess} = \max(0, \text{Used_Trucks} - \text{Limit}) \times \text{Penalty}$
- ACO (Overload Penalty): Massive multiplier for exceeding capacity (\$C\$).
 $\text{Fitness} = \text{Distance} \times 2$
- GA (Controlled Overload): Linear penalty for strategic overloads.
 $\text{Max_Load} = C \times 1.5 \text{ (Up to 150\%)}$

CONSTRAINTS



- The total demand of all customers on any given route must be less than or equal to the maximum weight/volume capacity of the vehicle.
- The three selected routes must maintain a total distance within 5% of the best-found solution, whilst exhibiting significant topological variations in their customer-to vehicle assignments



FITNESS FUNCTION COMPUTATION



1. Base Distance (Core Objective)

- Sum of the total distance traveled.
- Goal: Minimize kilometers to reduce operational costs.

2. Penalty System (Soft Constraints) Mathematical penalties for routes that violate capacity or fleet rules:

- PSO: Quadratic penalty for using excess vehicles.
- ACO: Massive multiplier (Distance x 2) for overloading trucks.
- GA: Linear penalty for strategic overloads (max 150%).

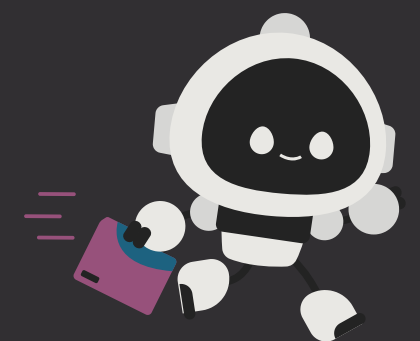
3. Quality Filter (The Archive)

- Once the final score (Distance + Penalties) is calculated, the route is only saved if it is excellent.
- Condition: Must be within a 5% gap compared to the optimal solution.

METAHEURISTICS USED



- Particle Swarm Optimization: we developed a multi-stage architecture to overcome dimensionality issues and meet multimodal requirements.
- Ant Colony Optimization: we extended the standard architecture with specialized multimodal memories and sequential restarts to prevent the colony from converging on a single, dominant trail
- Genetic Algorithm, our GA architecture was heavily upgraded into a Memetic Algorithm (integrating localized 2-opt repair) and reinforced with explicit niching mechanisms (deterministic crowding, sequential tabu memory)





PSO

PSO REPRESENTATION



A particle's position is defined as a permutation of customers (a "Giant Tour"). The movement logic is adapted to this discrete space by redefining "Velocity" as a sequence of transposition operators (swaps).

PROBABILISTIC MOVEMENT



To move a particle, we calculate the swap sequences required to reach pBest and gBest. Unlike the continuous version, the parameters w , c_1 , and c_2 are interpreted as probabilities (range $[0, 1]$). These probabilities dictate whether a specific swap from the sequence is actually performed.



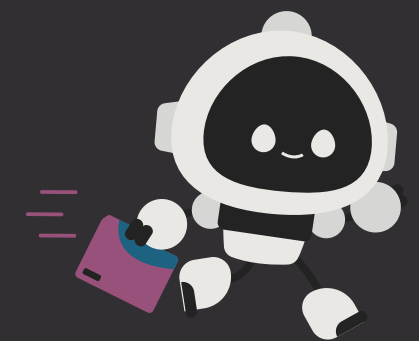
CONSTRAINT HANDLING



Managing the vehicle capacity constraint depends on the algorithmic phase:

Standard Decoding: For full-map approaches or within isolated clusters, the permutation is decoded using a splitting procedure. It iterates through the tour and inserts a return to the depot whenever adding the next customer would exceed the maximum vehicle capacity C .

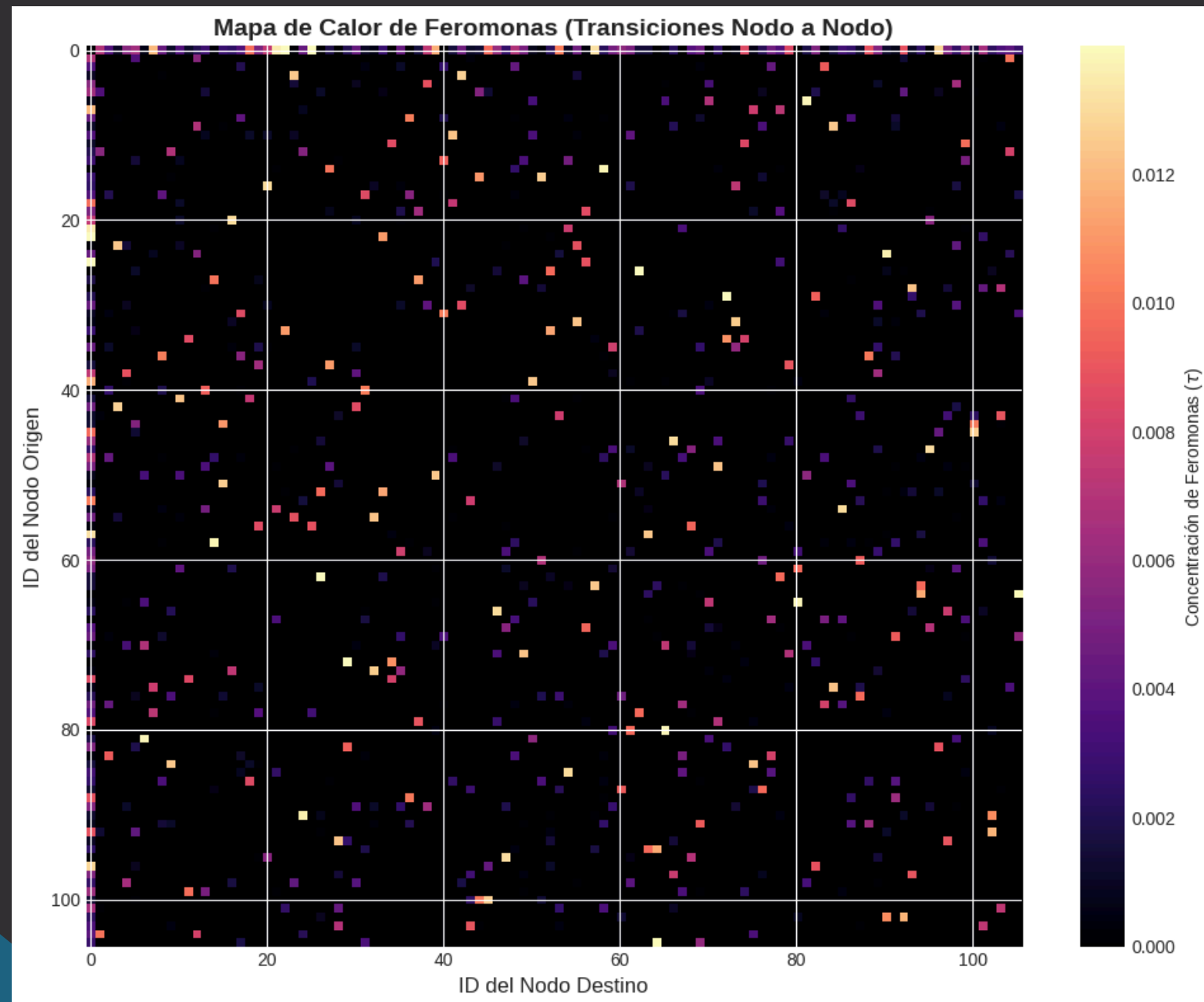
Clustering Consolidation: In Phase 2, independent feasible routes from each cluster are merged to reduce the total fleet size.





ACO

PHEROMONES



- Y axis: source node
- X axis destination node

HEURISTIC



An ant selects the next customer j from its current position i probabilistically, proportional to the equation:

$$[\tau_{ij}]^{\alpha} [\eta_{ij}]^{\beta}$$

where τ_{ij} is the pheromone level and $\eta_{ij} = 1/d_{ij}$ is the heuristic visibility.

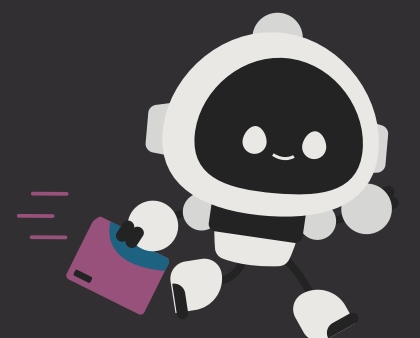
CONSTRAINT HANDLING



To manage the vehicle capacity constraint C , the ACO employs two distinct evaluated strategies:

Feasibility (Dynamic): The feasible neighborhood is strictly filtered during construction. An ant can only transition to customers whose demand fits the remaining truck capacity; otherwise, it is forced to return to the depot (node 0) to reset its load.

Penalty (Soft Constraint): Ants construct routes freely, but upon completion, a massive mathematical penalty is applied to the objective function if the total route violates fleet or capacity limits. This forces the colony to learn the boundaries through pheromone evolution rather than hard-coded rules.





GA

REPRESENTATION & OPERATORS

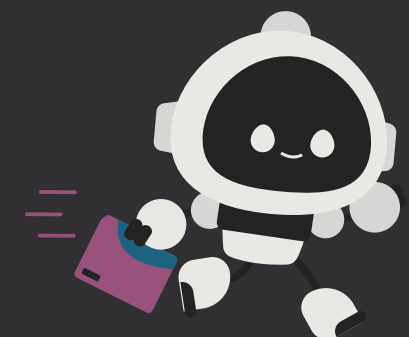


- Chromosome: A solution is codified as a pure permutation of all customer IDs (a "Giant Tour").
- Recombination (OX1): Order Crossover is used to inherit structural sub-routes from parents, ensuring no customers are duplicated or omitted.
- Mutation (Inversion): Sub-sequences of the tour are reversed. This acts as a high-energy disruption to untangle crossed edges and escape local optima.

CONSTRAINT HANDLING



The Giant Tour is decoded into valid routes using a Greedy Splitter approach. The decoder traverses the chromosome and automatically inserts a return to the depot whenever adding the next customer would exceed the maximum vehicle capacity C , guaranteeing 100% mathematically feasible solutions.

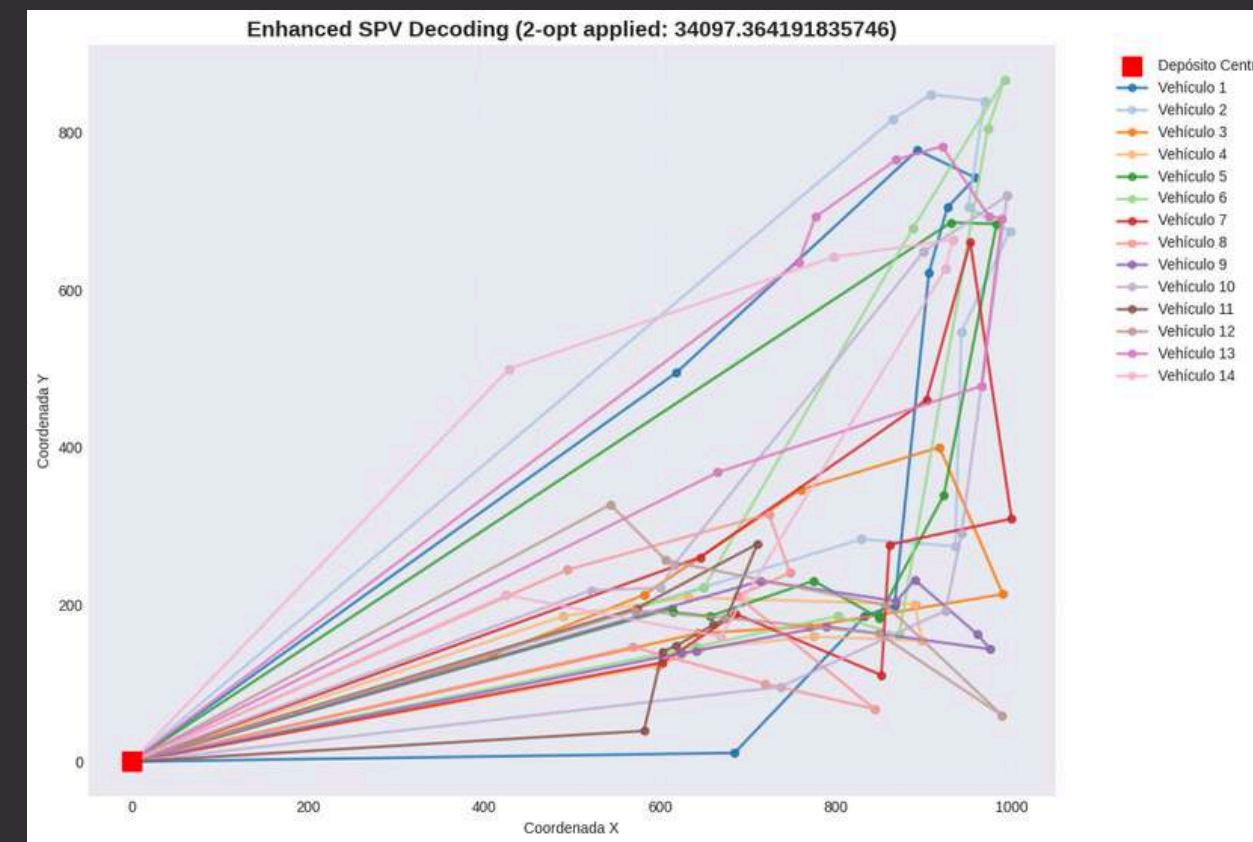
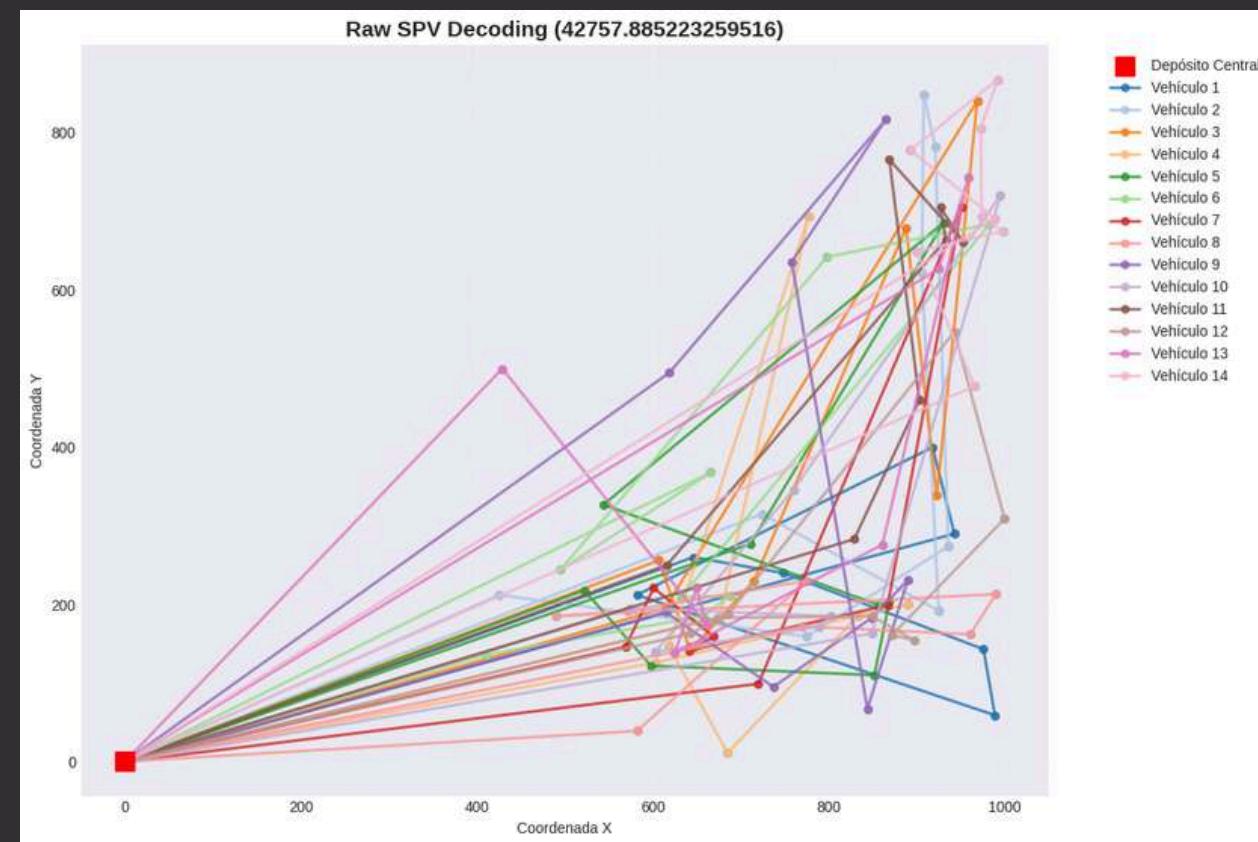




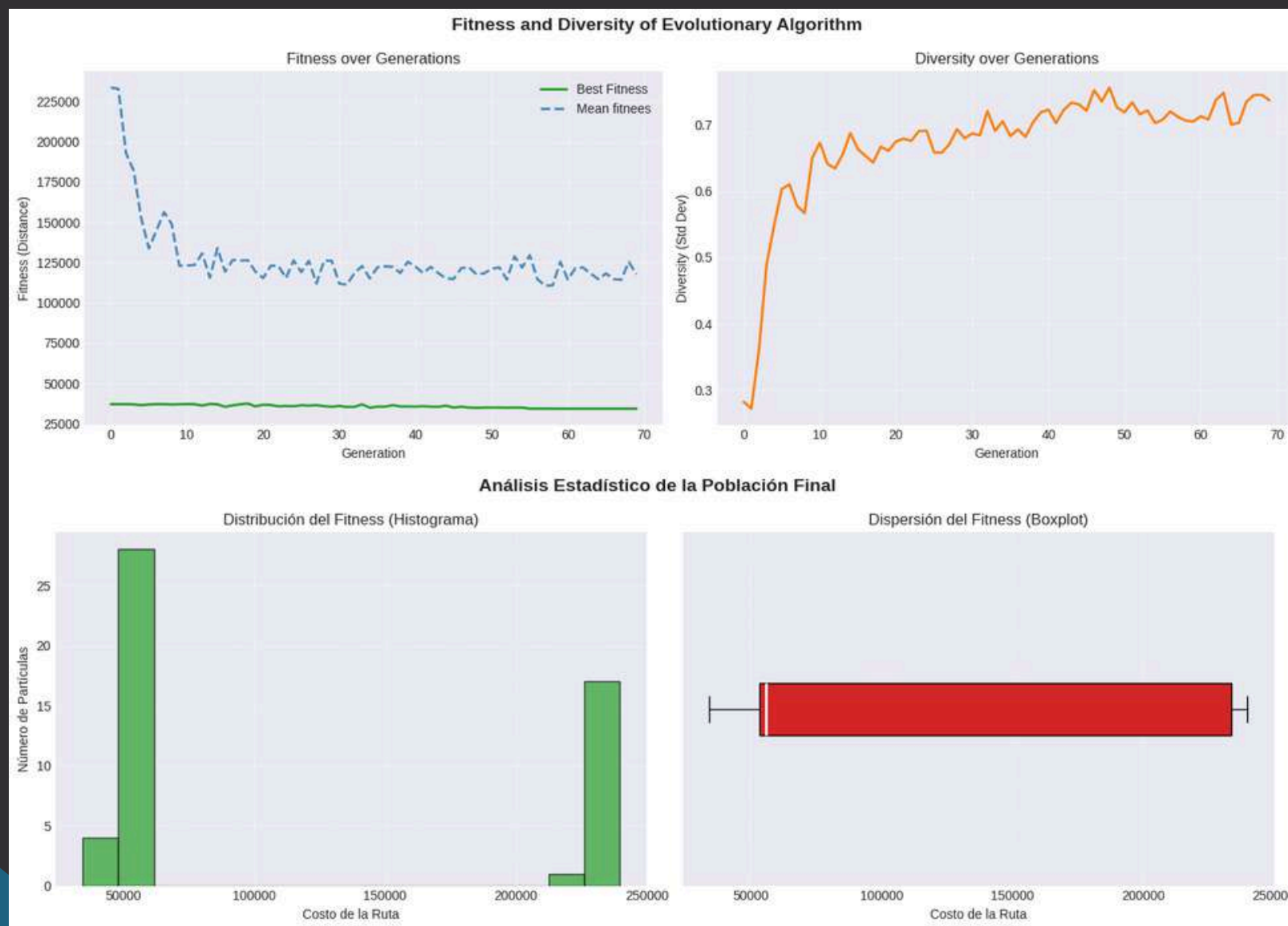
EXPERIMENTS

PSO PARAMETER SETTINGS

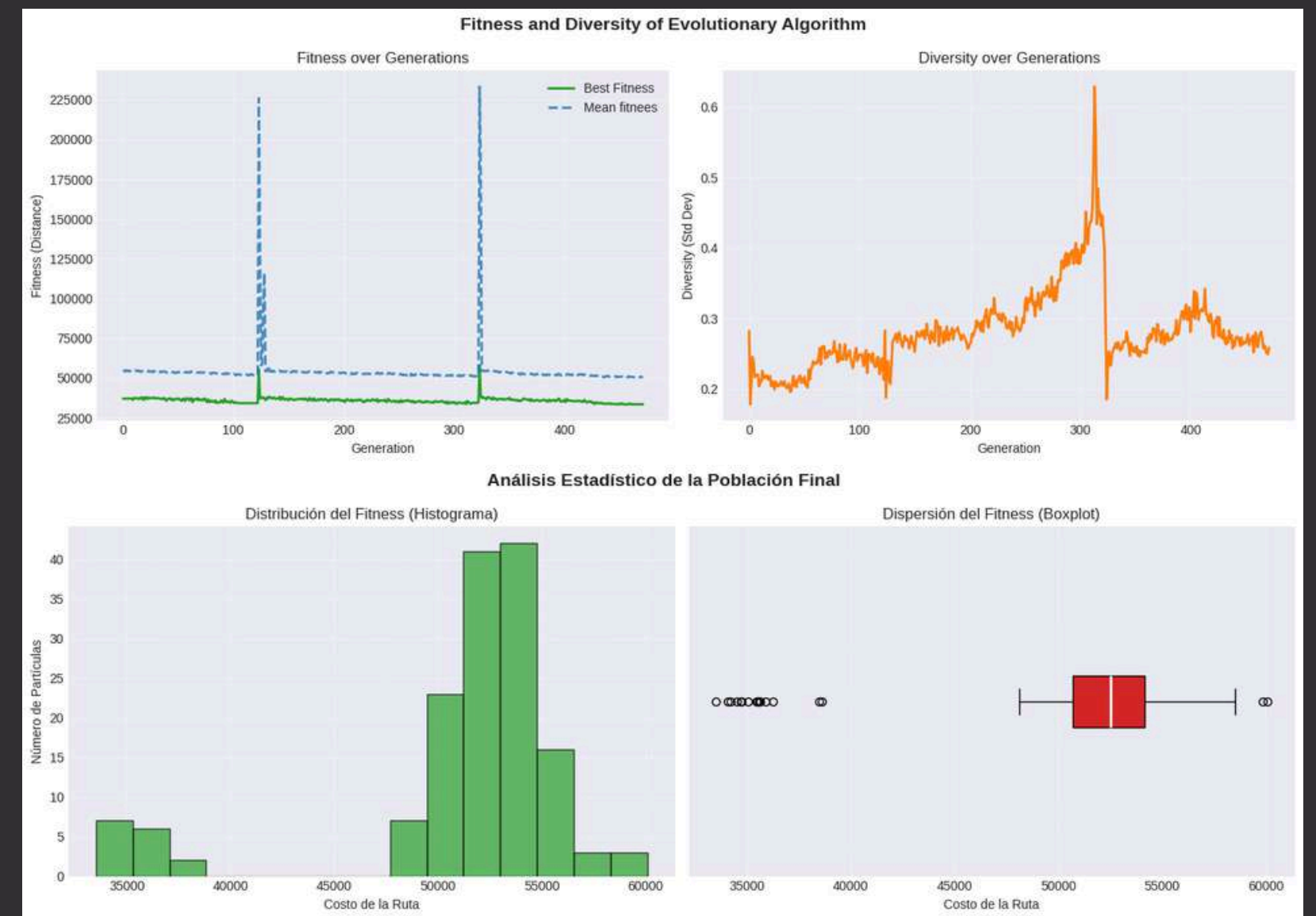
Phase 1: Baseline Evaluation and the Necessity of Local Search (2-opt)



PSO PARAMETER SETTINGS

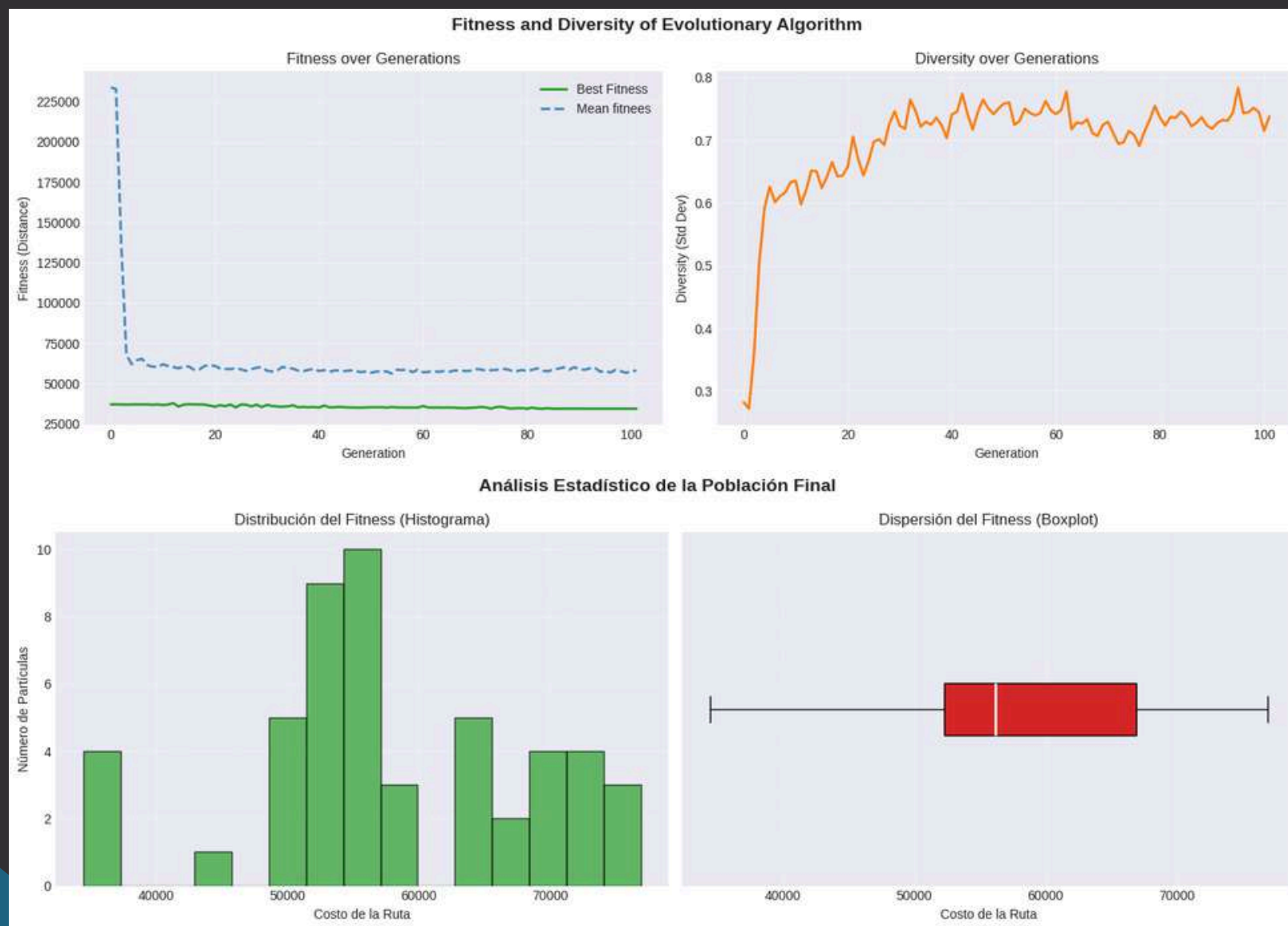


Clearing

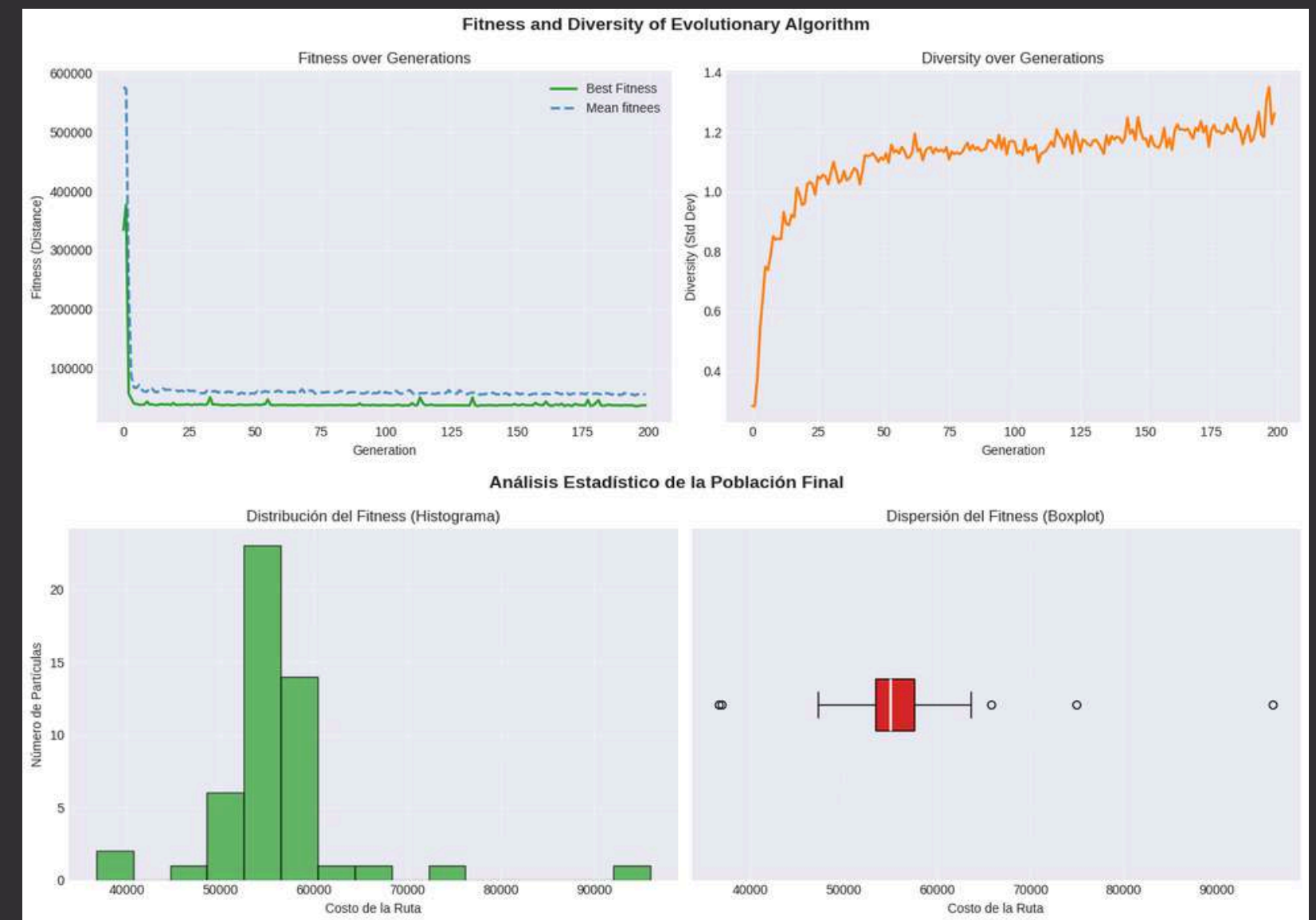


Sequential Niching

PSO PARAMETER SETTINGS



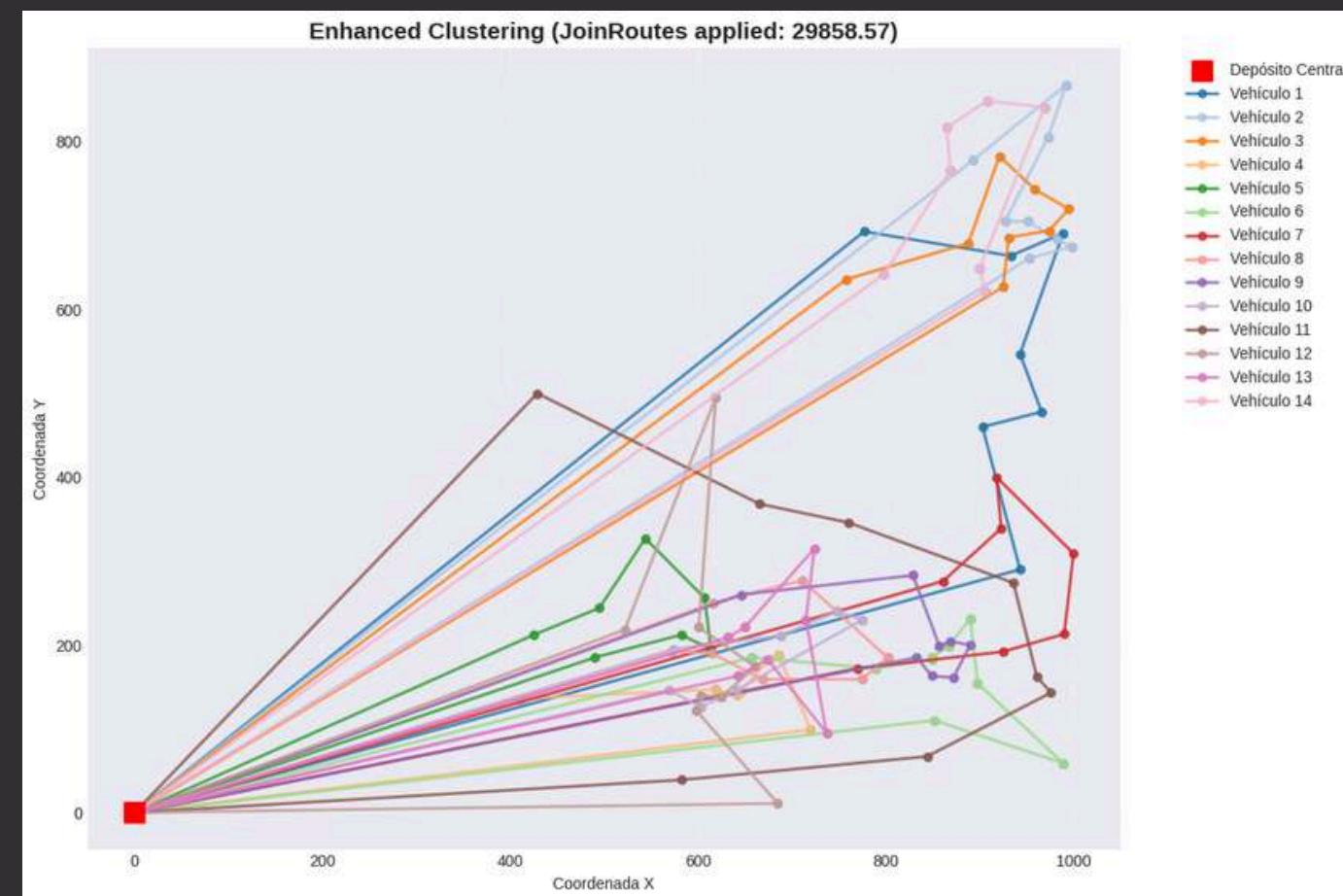
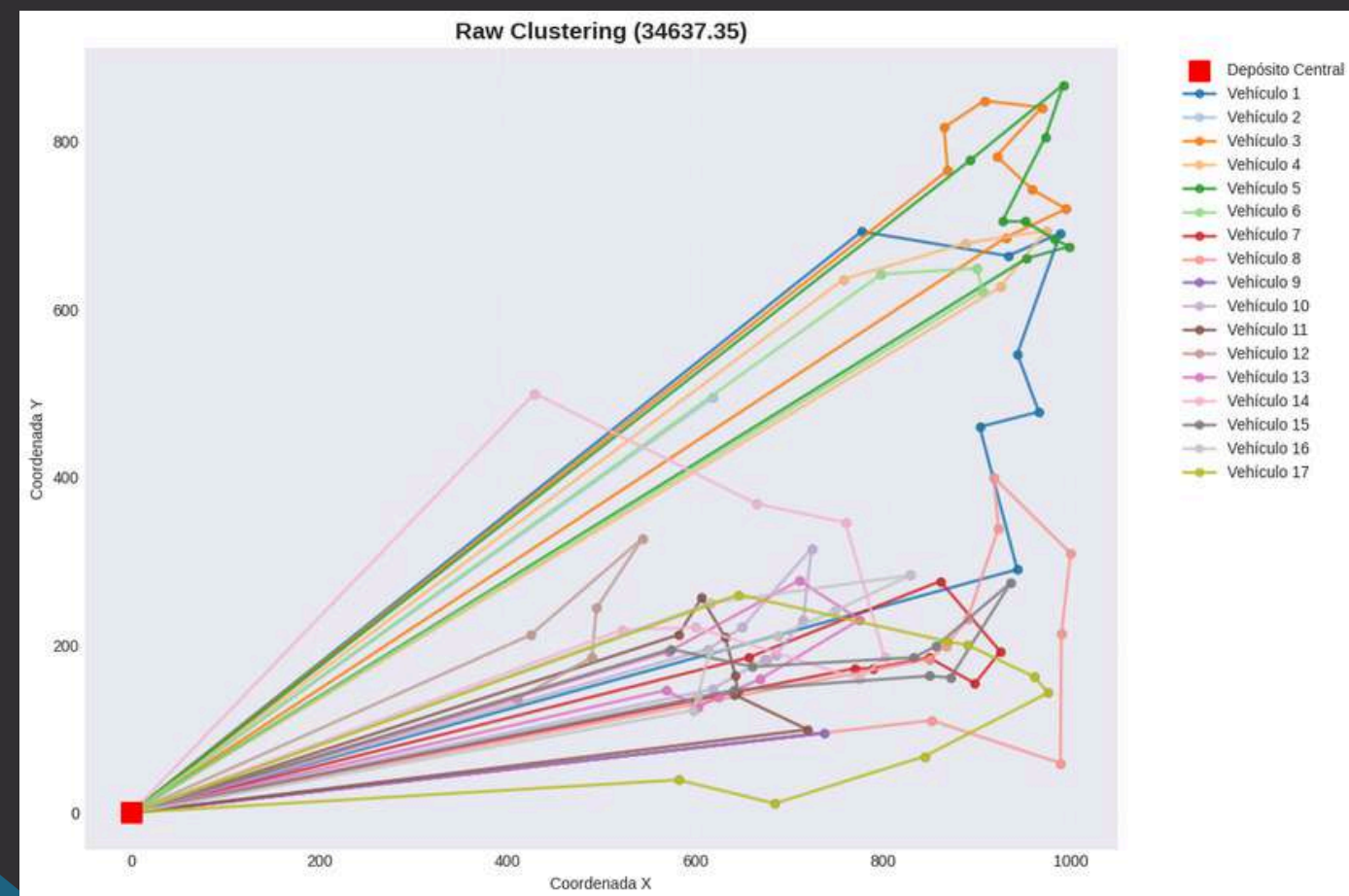
Species Competition



Fitness Sharing

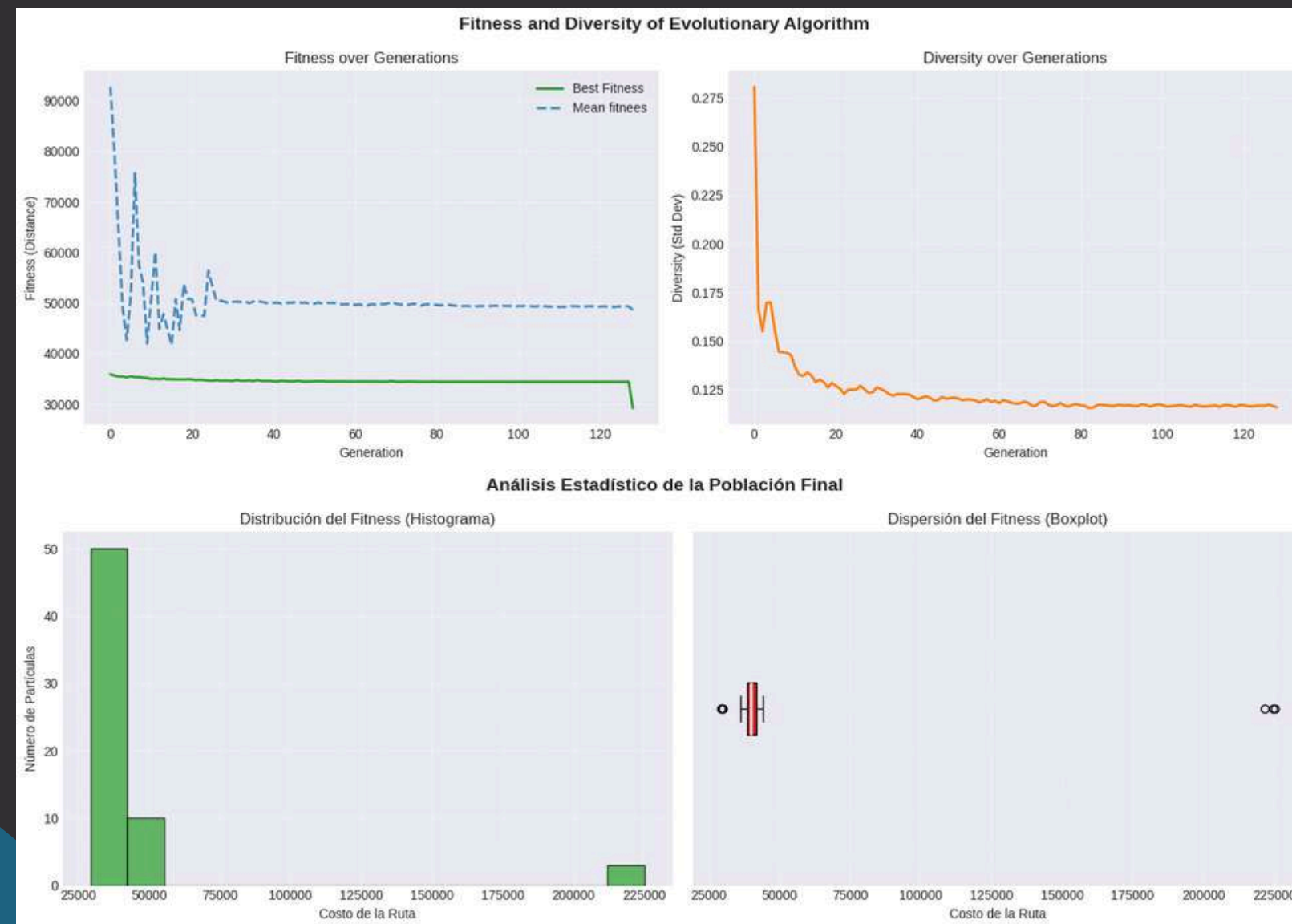
PSO PARAMETER SETTINGS

Phase 2: Clustering Architecture and the "JoinRoutes" Consolidation

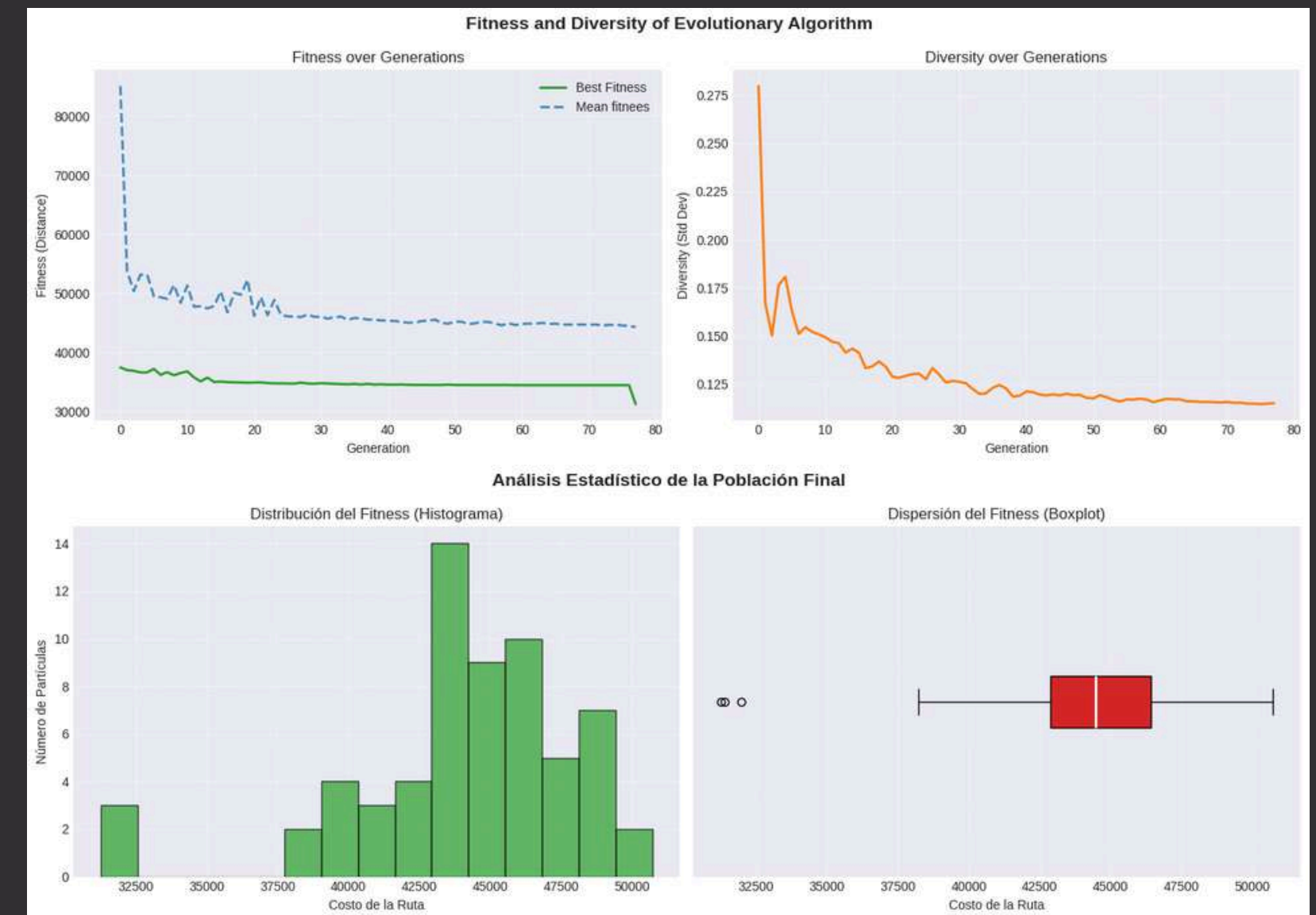


PSO PARAMETER SETTINGS

Phase 3: Evaluation of Clustering Strategies

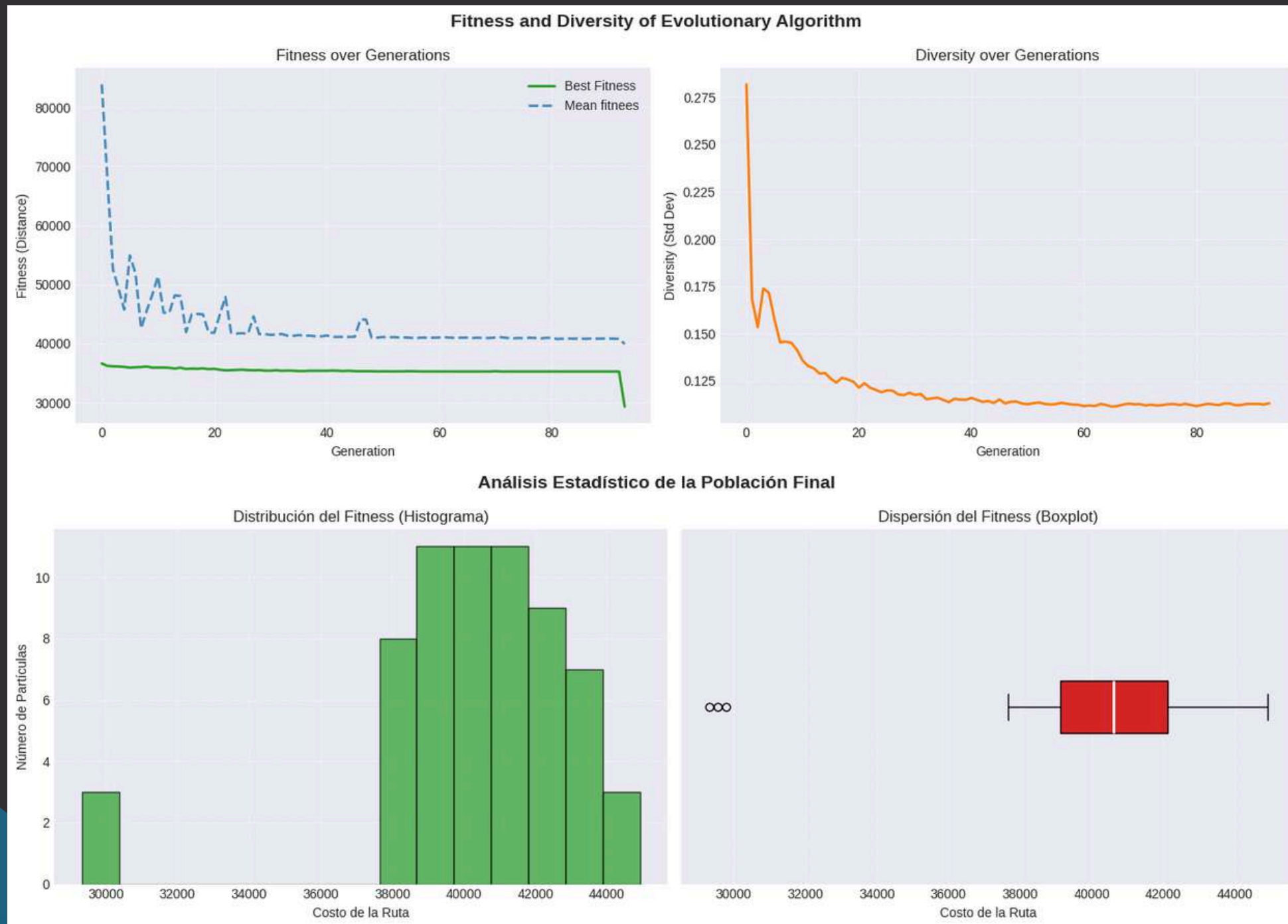


Normal Clustering



Angular Clustering

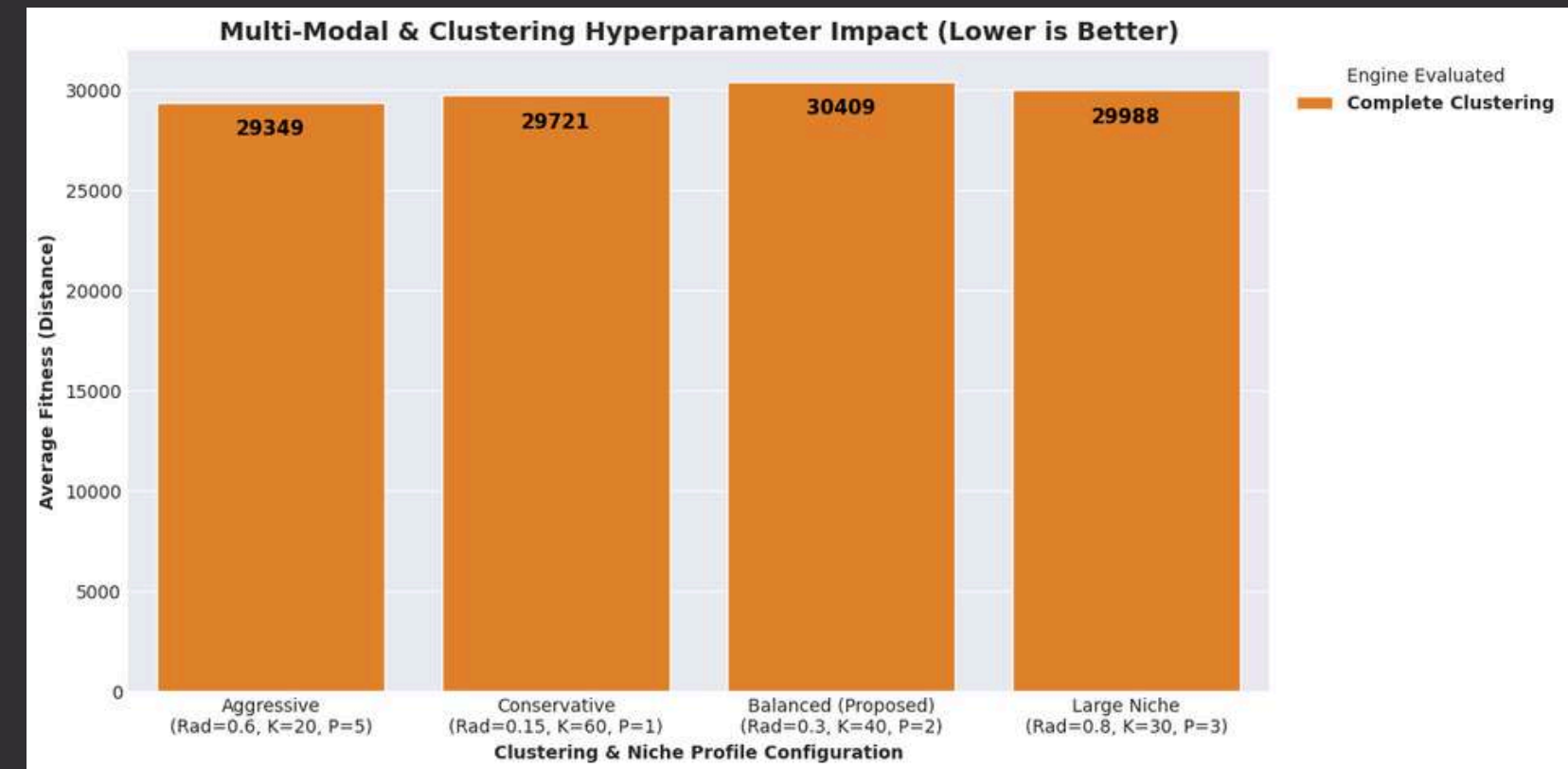
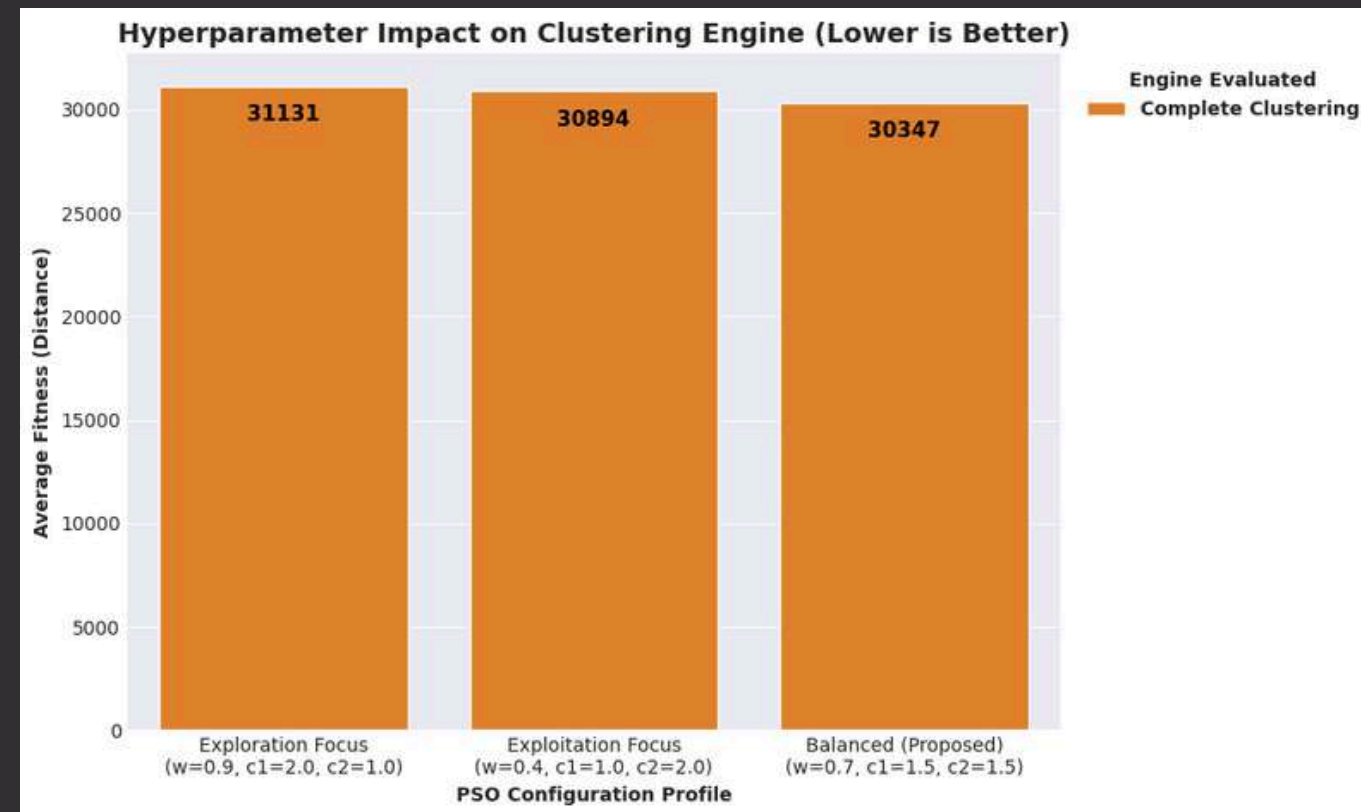
PSO PARAMETER SETTINGS



Complete Clustering (Hybrid)

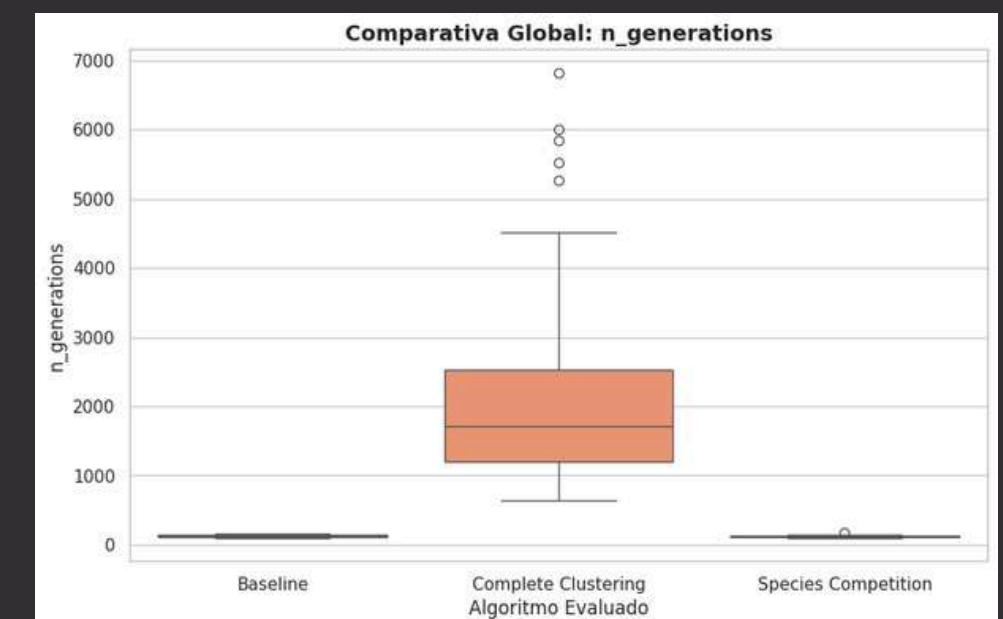
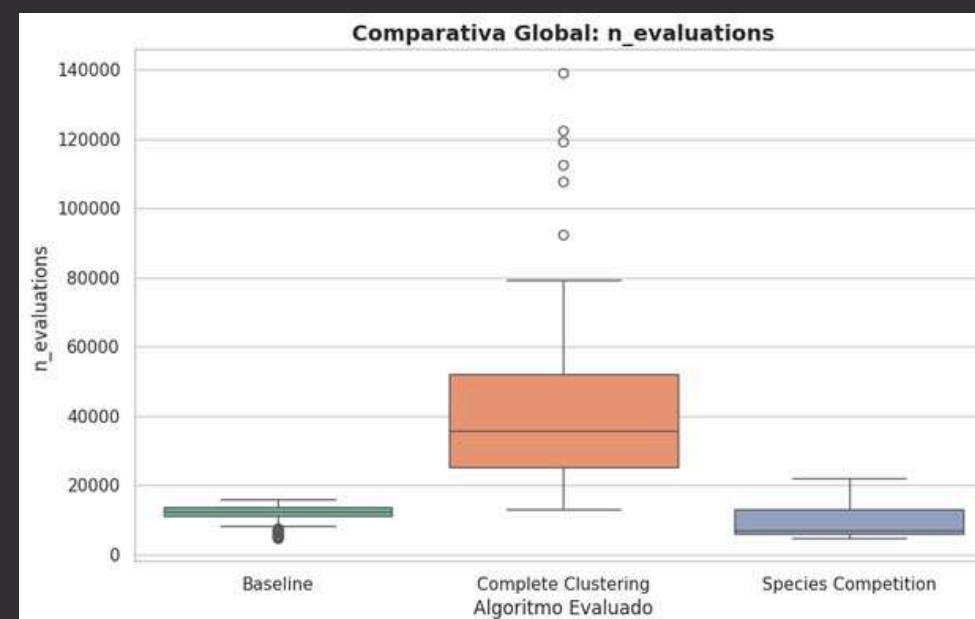
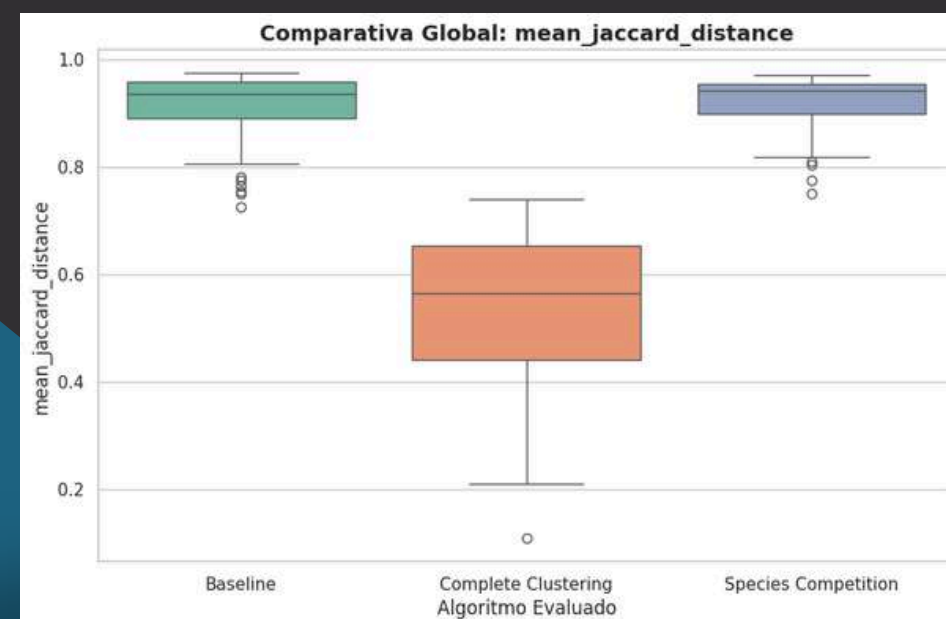
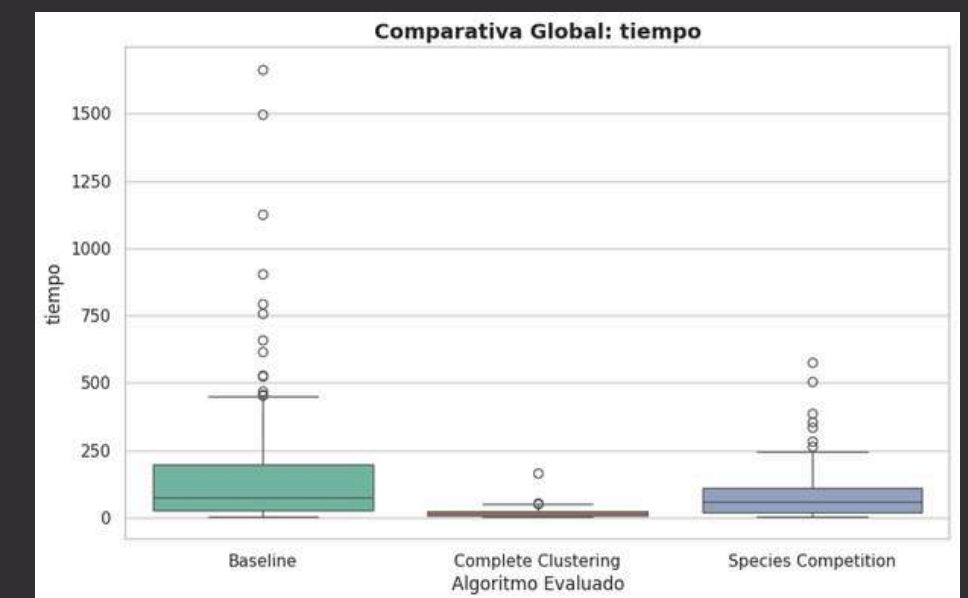
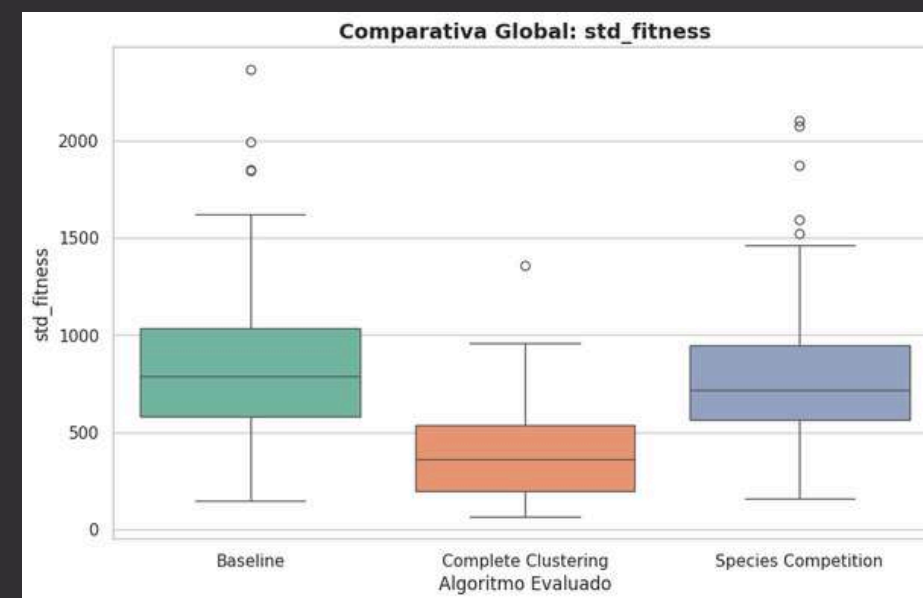
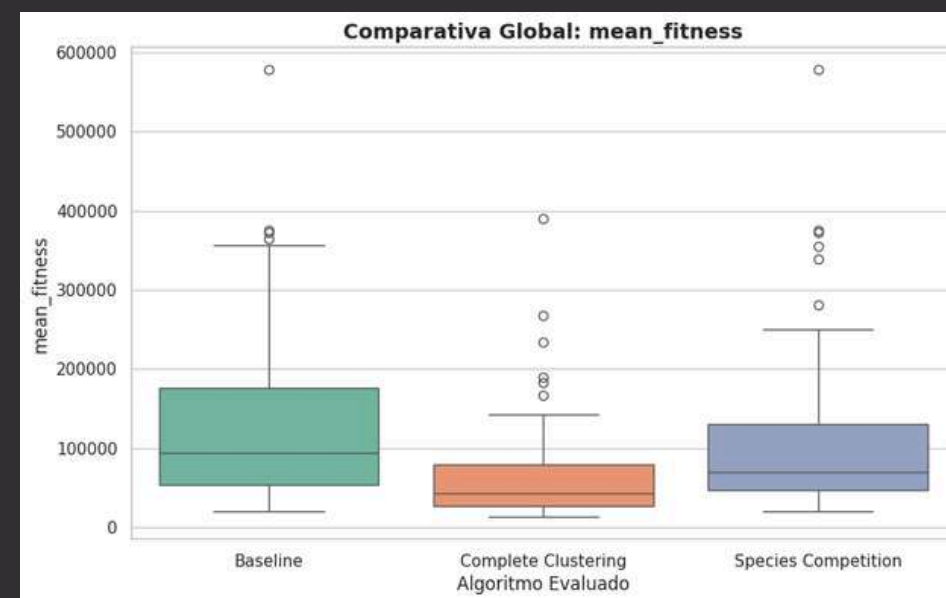
PSO PARAMETER SETTINGS

Phase 4: Hyperparameter Tuning



PSO PARAMETER SETTINGS

Phase 5: Statistical Significance Testing



PSO PARAMETER SETTINGS

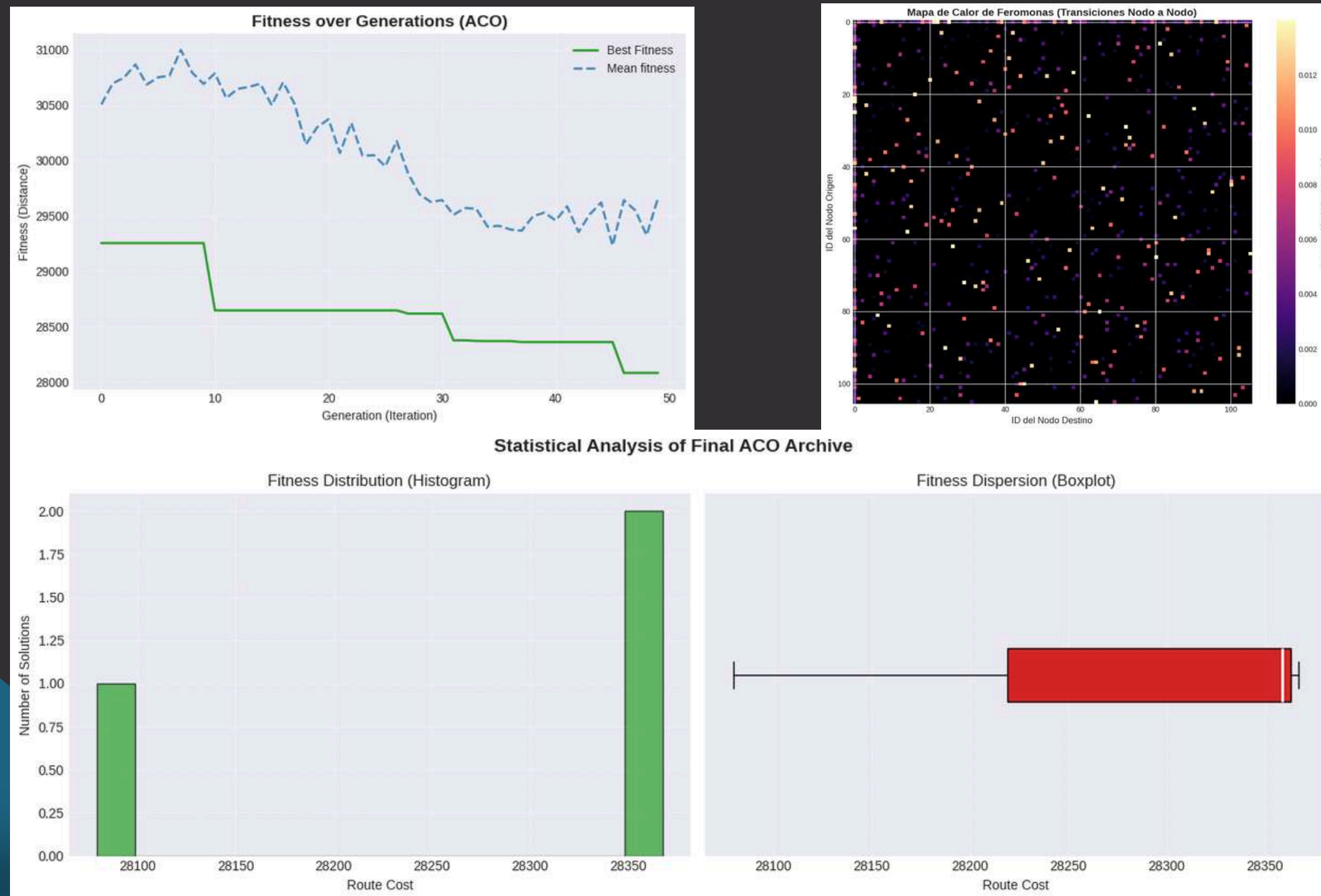


Final Verdict: the integration of structural Complete Clustering and local heuristics transforms the PSO from a generic continuous optimizer into a specialized CVRP solver. The results unequivocally demonstrate that structural hybridization is the only viable path for resolving complex multimodal logistical constraints efficiently.

Feature	Baseline PSO	Species Competition	Complete Clustering
Solution Quality	Moderate	Moderate	Excellent
Execution Speed	Slow	Slow	Very Fast
Stability (Std)	Low	Low	High
Diversity	High	High	Moderate
Implementation	Simple	Complex	Highly Complex

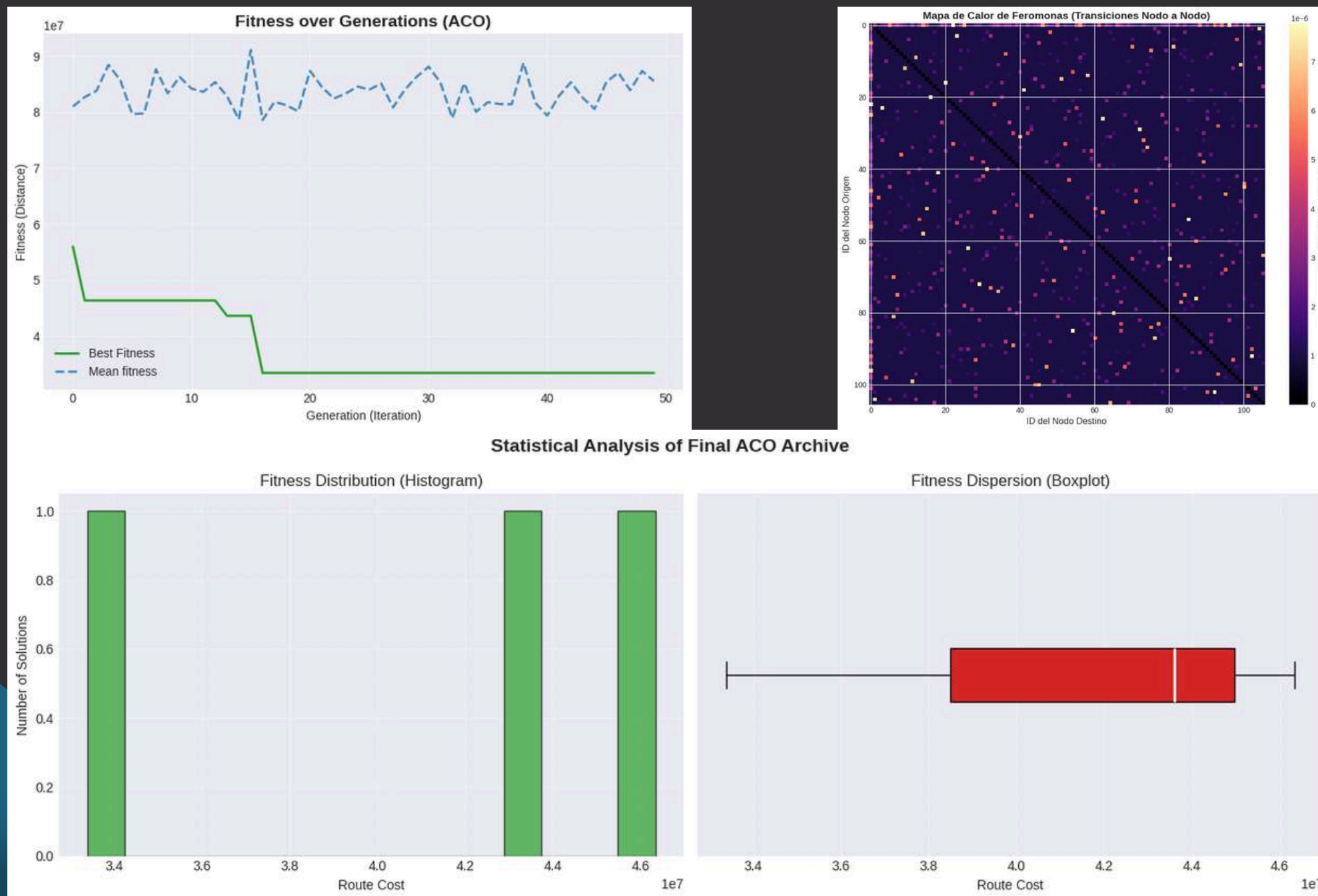
ACO PARAMETER SETTINGS

Phase 1: The Archive Method with Feasibility Constraints



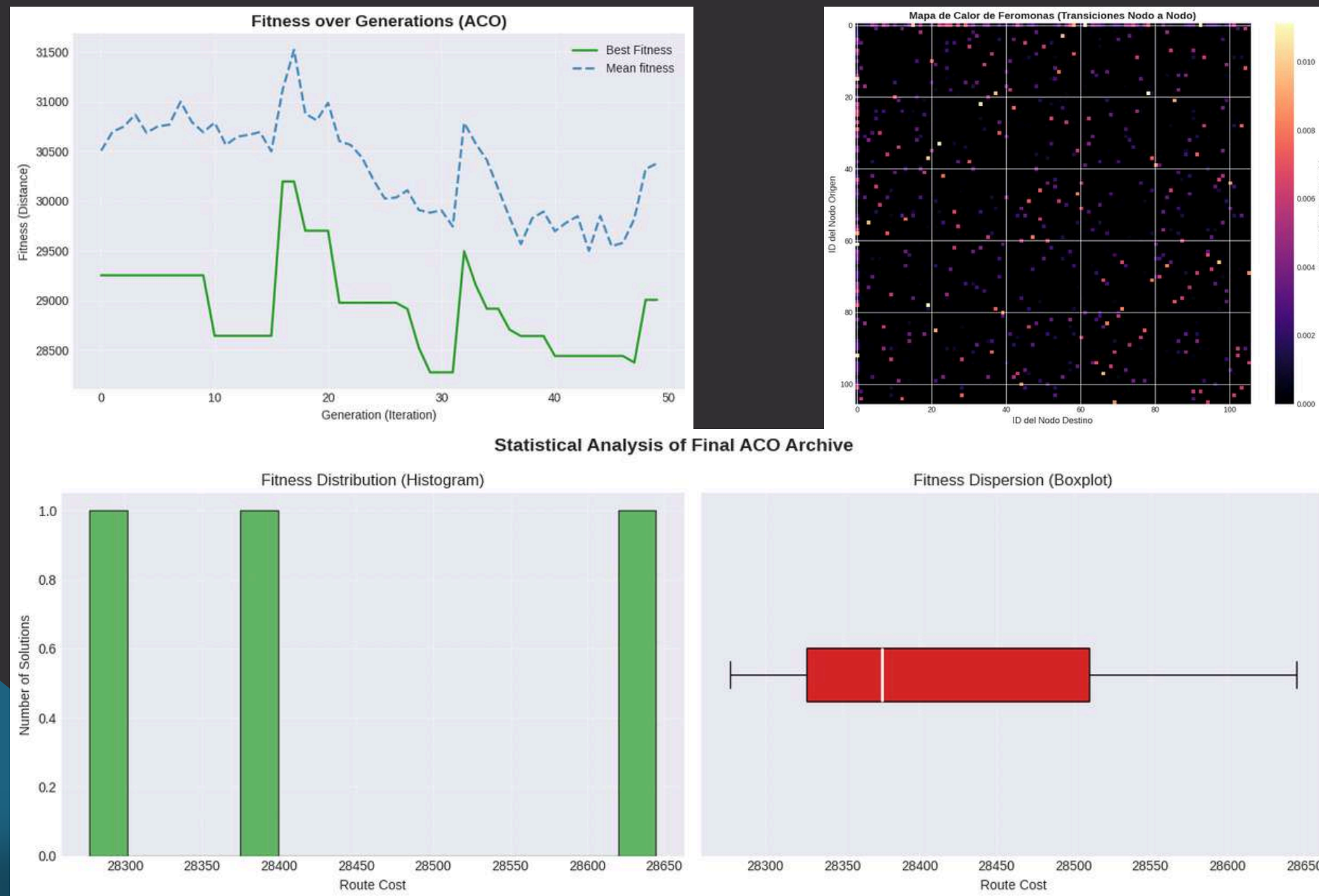
ACO PARAMETER SETTINGS

Phase 2: The Archive Method with Adaptive Penalty Constraints



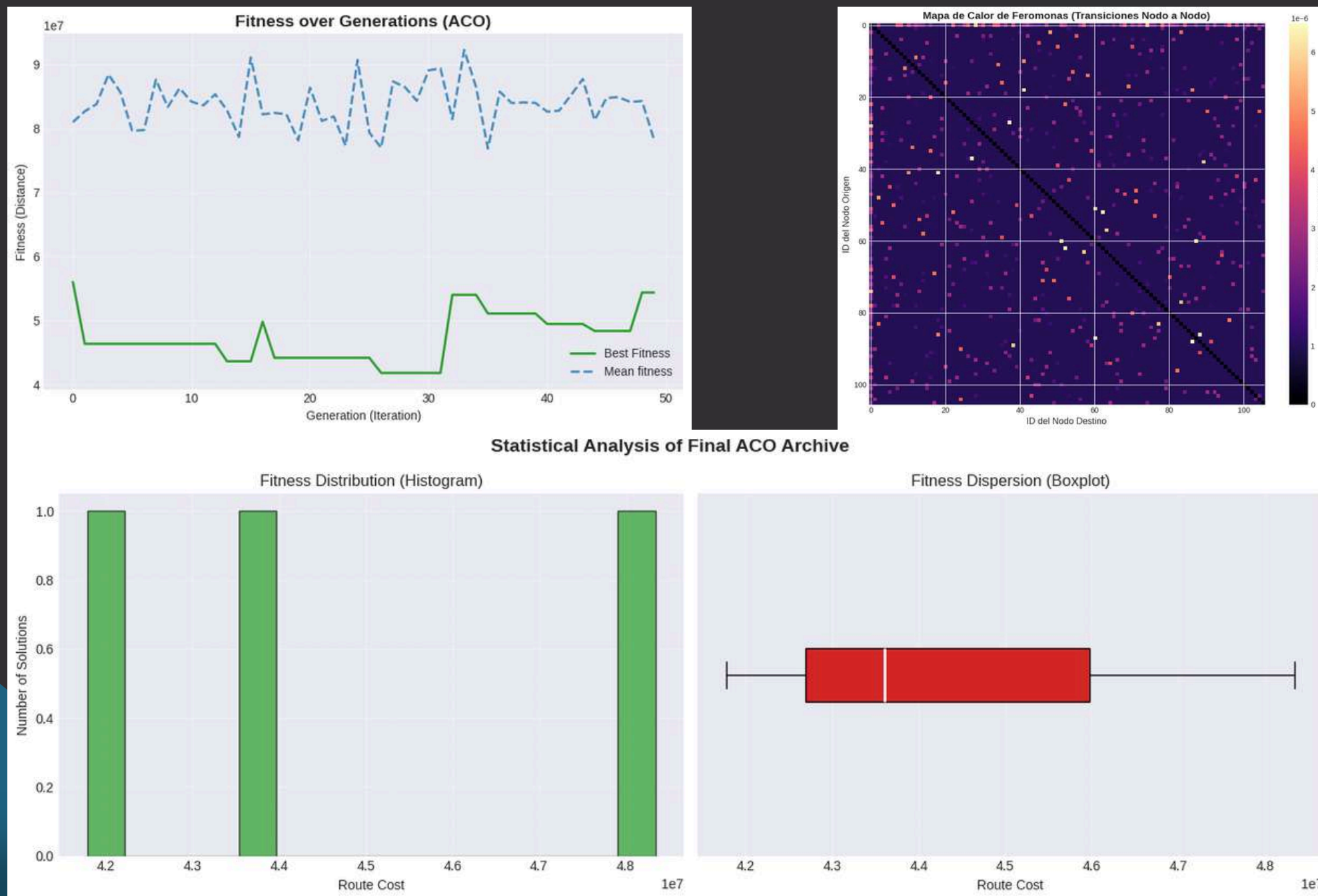
ACO PARAMETER SETTINGS

Phase 3: The Sequential Method with Feasibility Constraints



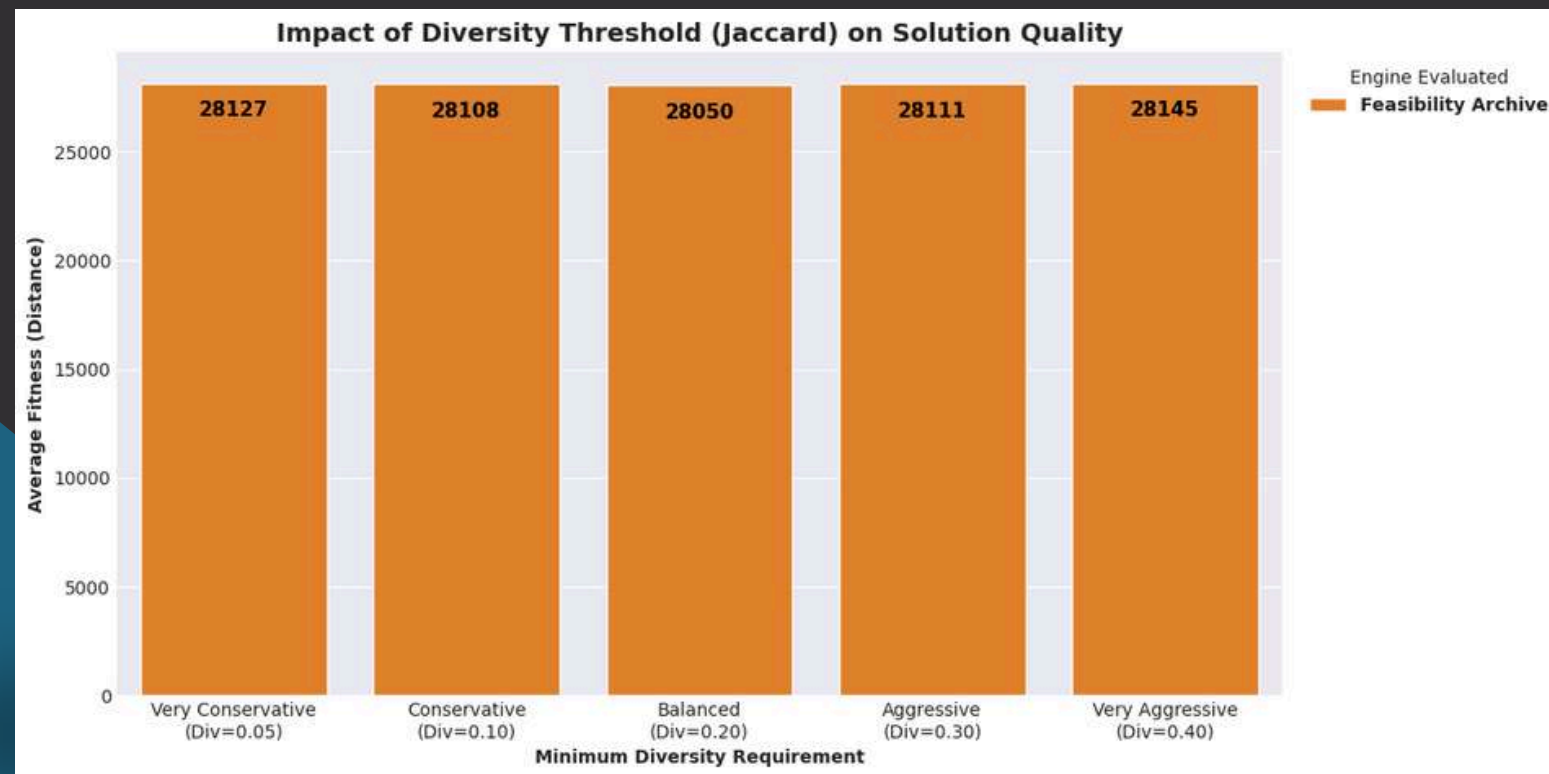
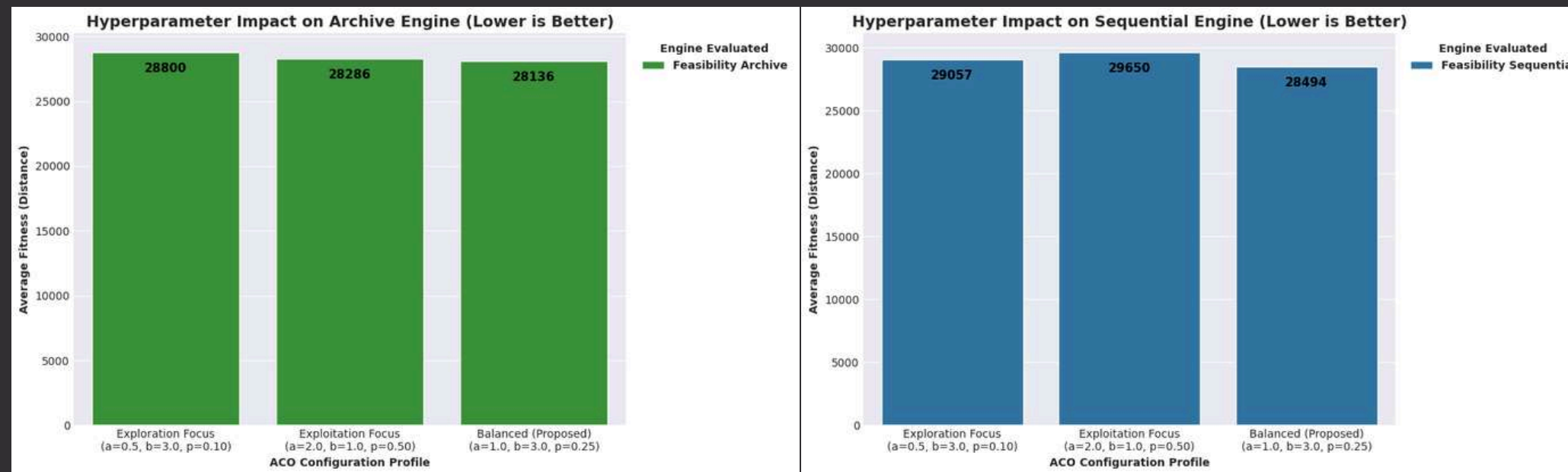
ACO PARAMETER SETTINGS

Phase 4: The Sequential Method with Adaptive Penalty Constraints



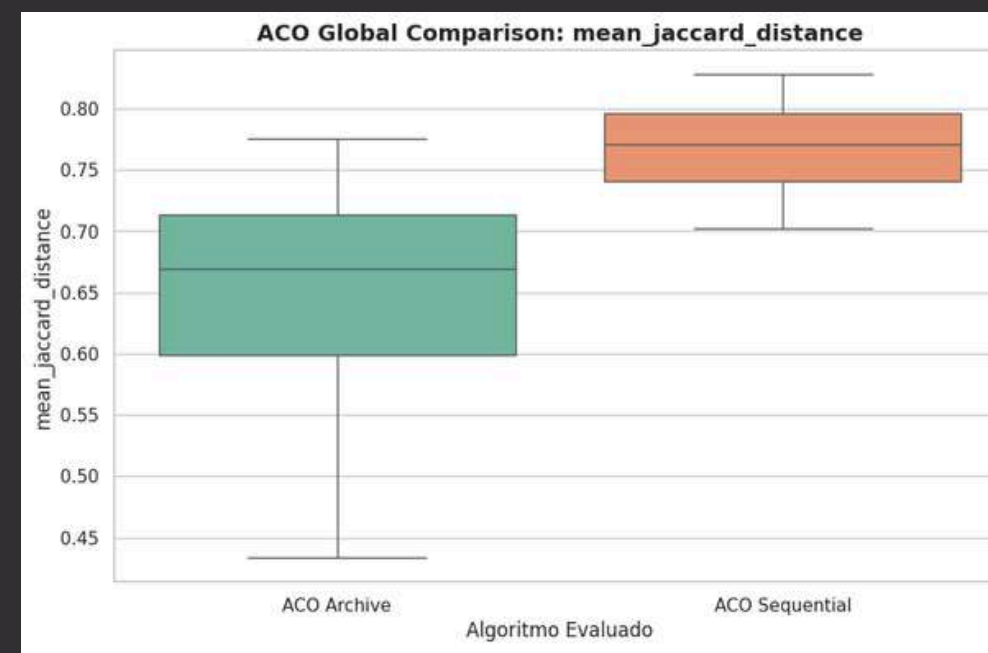
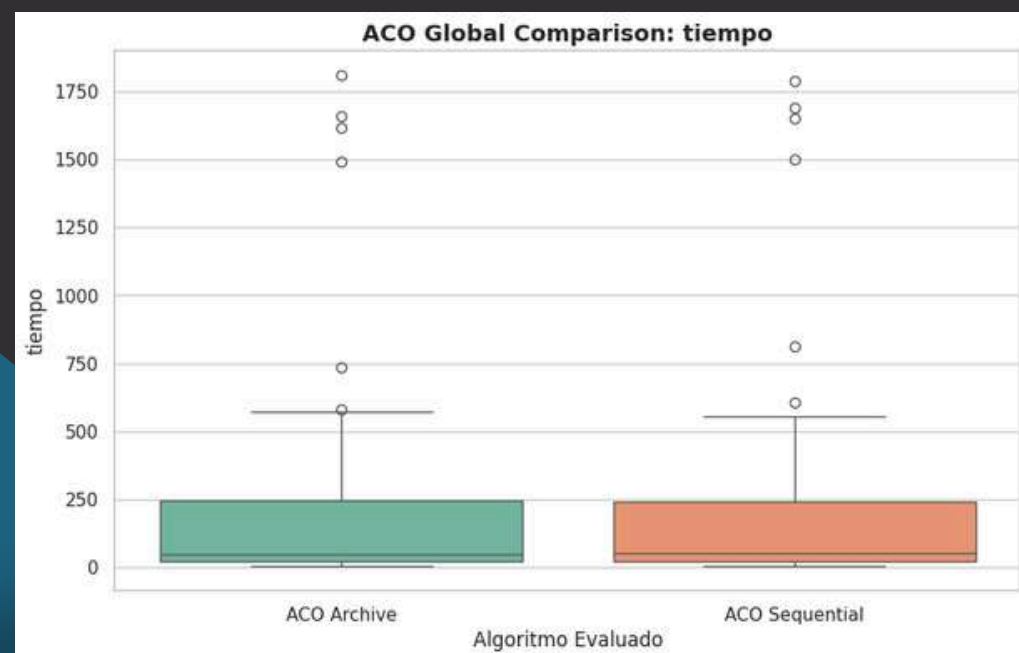
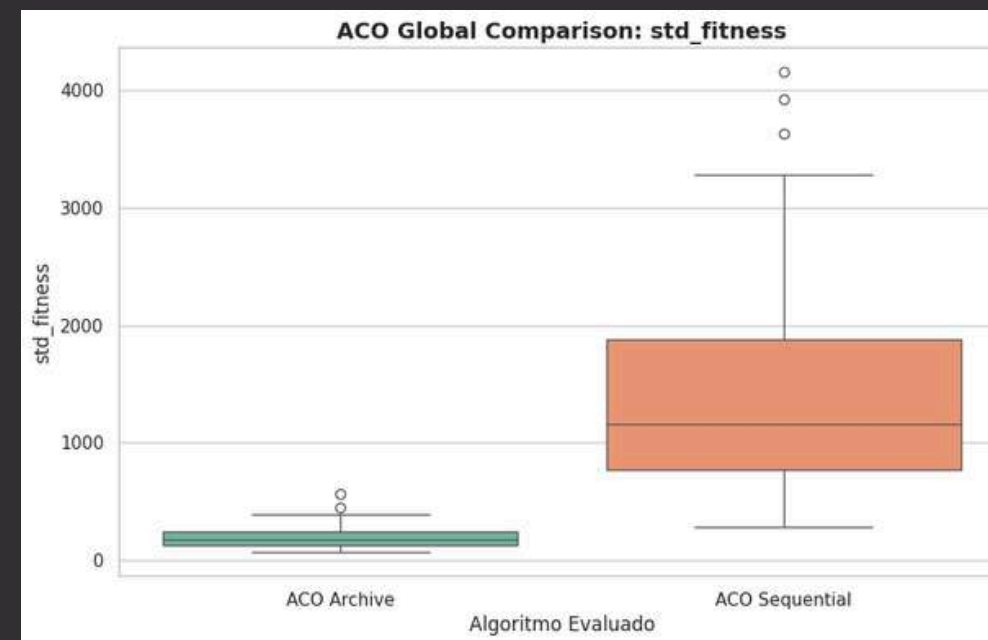
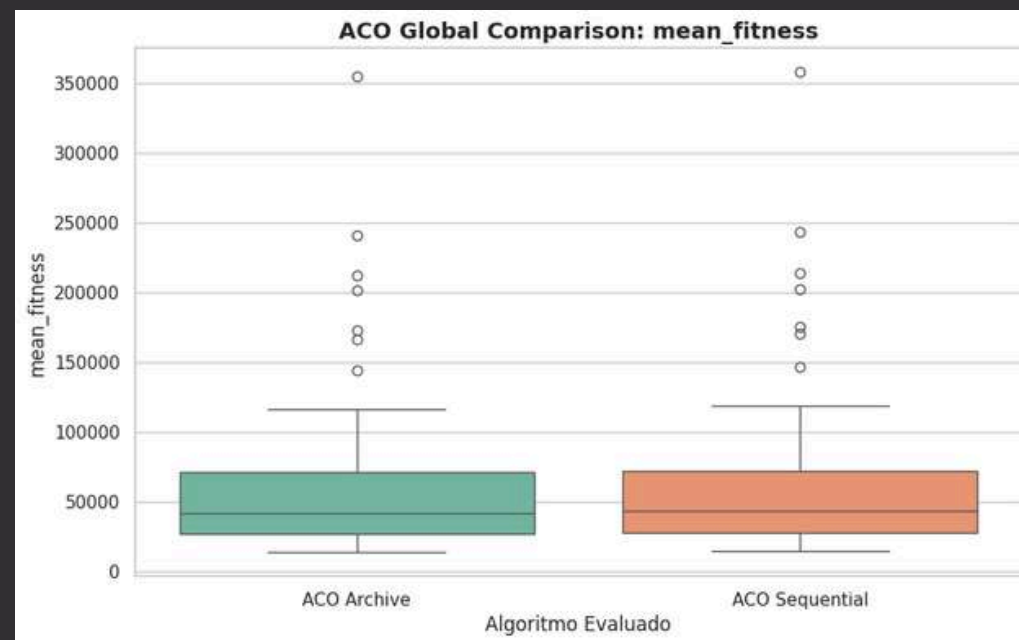
ACO PARAMETER SETTINGS

Phase 5: Hyperparameter Tuning



ACO PARAMETER SETTINGS

Phase 6: Statistical Significance Testing



ACO PARAMETER SETTINGS



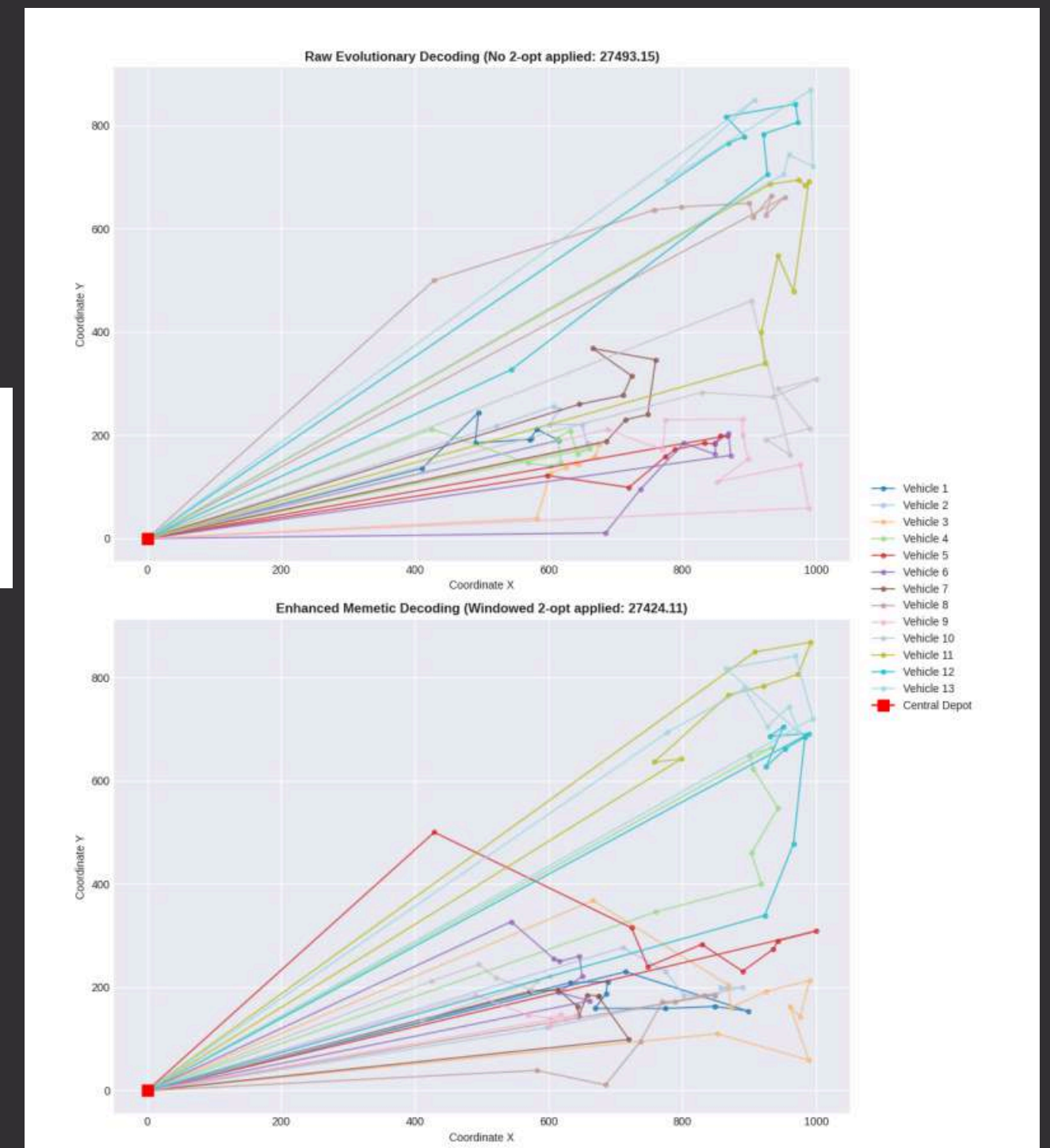
Final Verdict: The integration of a feasibility-based passive archive transforms the ACO into a highly specialized and robust CVRP solver. The results unequivocally show that respecting the natural evolution of pheromone trails while applying strict capacity boundaries is the mathematically superior path for solving complex multimodal logistical constraints

Feature	Sequential ACO	Archive ACO
Solution Quality	Good	Excellent
Execution Speed	Fast (Tie)	Fast (Tie)
Stability (Std)	Moderate	High
Diversity (Jaccard)	Highest	Moderate
Swarm Learning	Active (Erasure)	Passive (Preserved)

GA PARAMETER SETTINGS

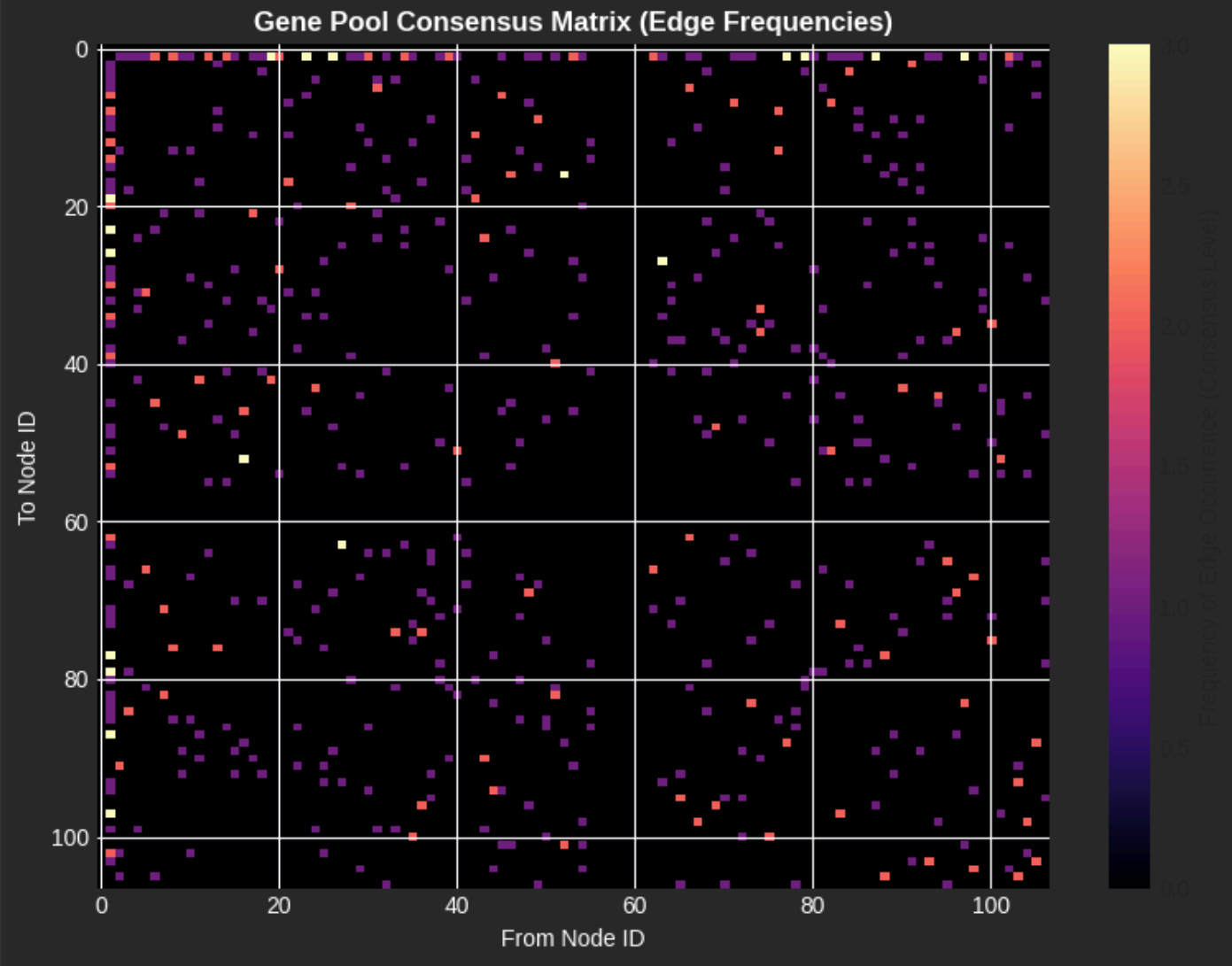
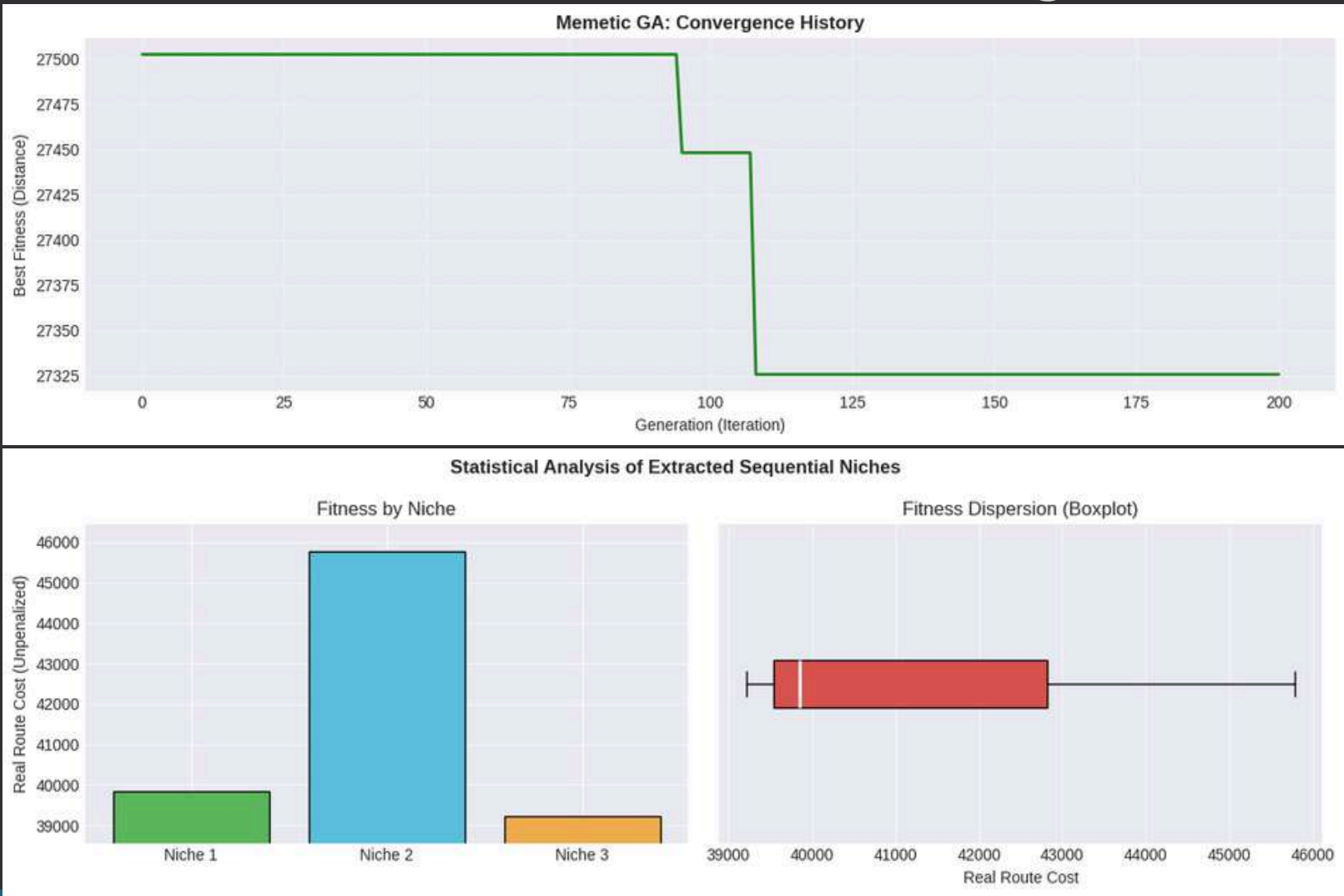
Phase 1: Pure GA vs. Memetic Analysis (The Engine)

Metric	Pure GA (Baseline)	Enhanced (Memetic GA)	Improvement
Total Fitness	27,493.15	27,424.11	~69.04 Reduction
Execution Time	54.10s	64.37s	+10.27s Overhead



GA PARAMETER SETTINGS

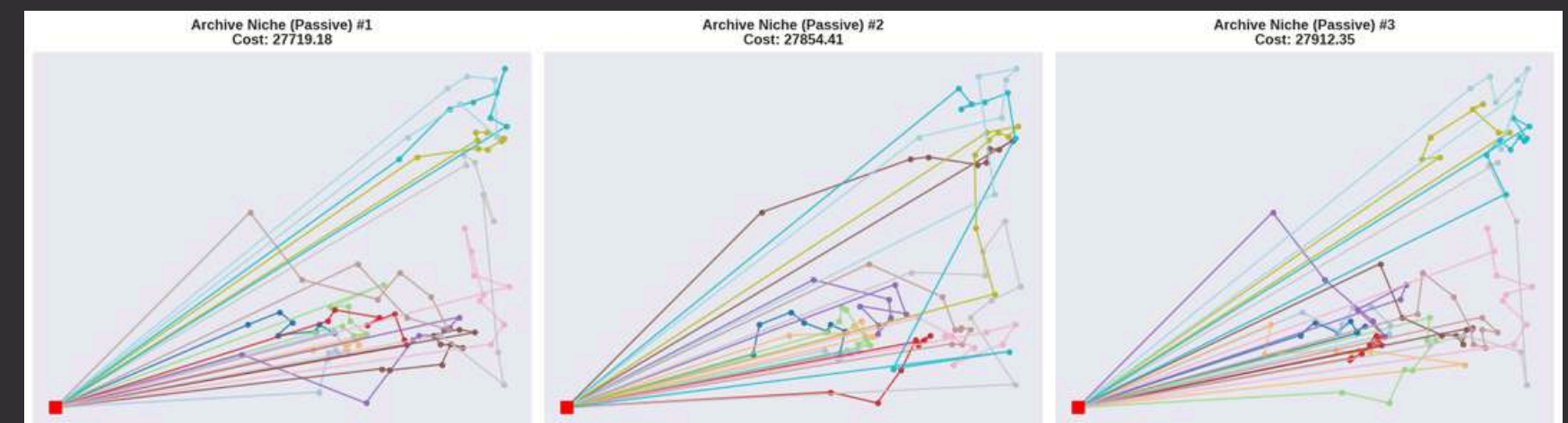
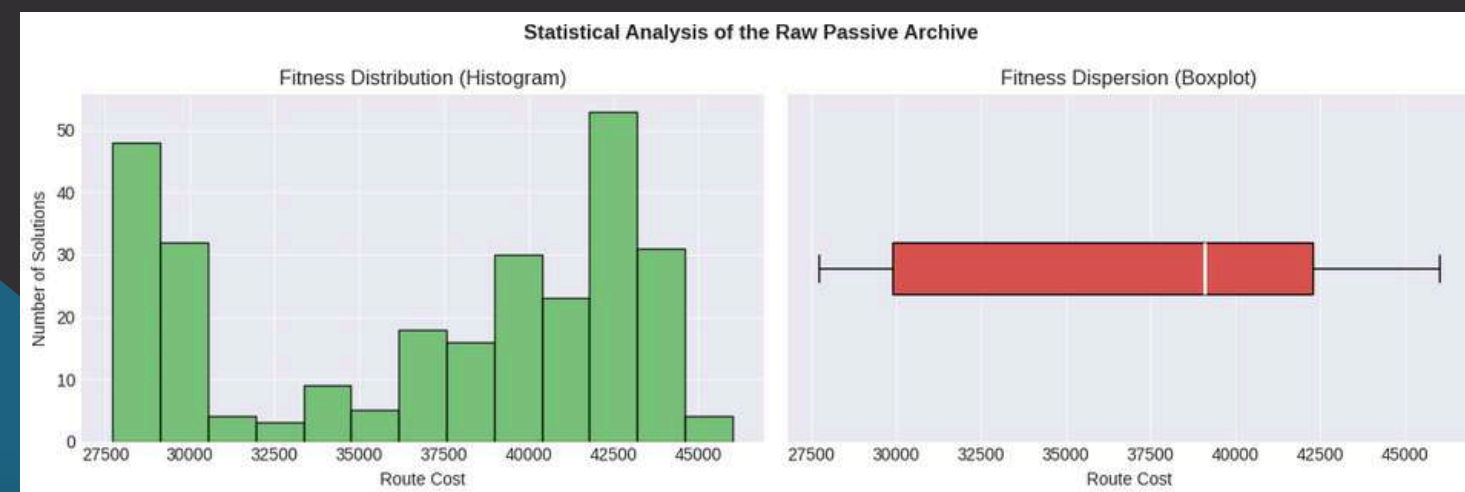
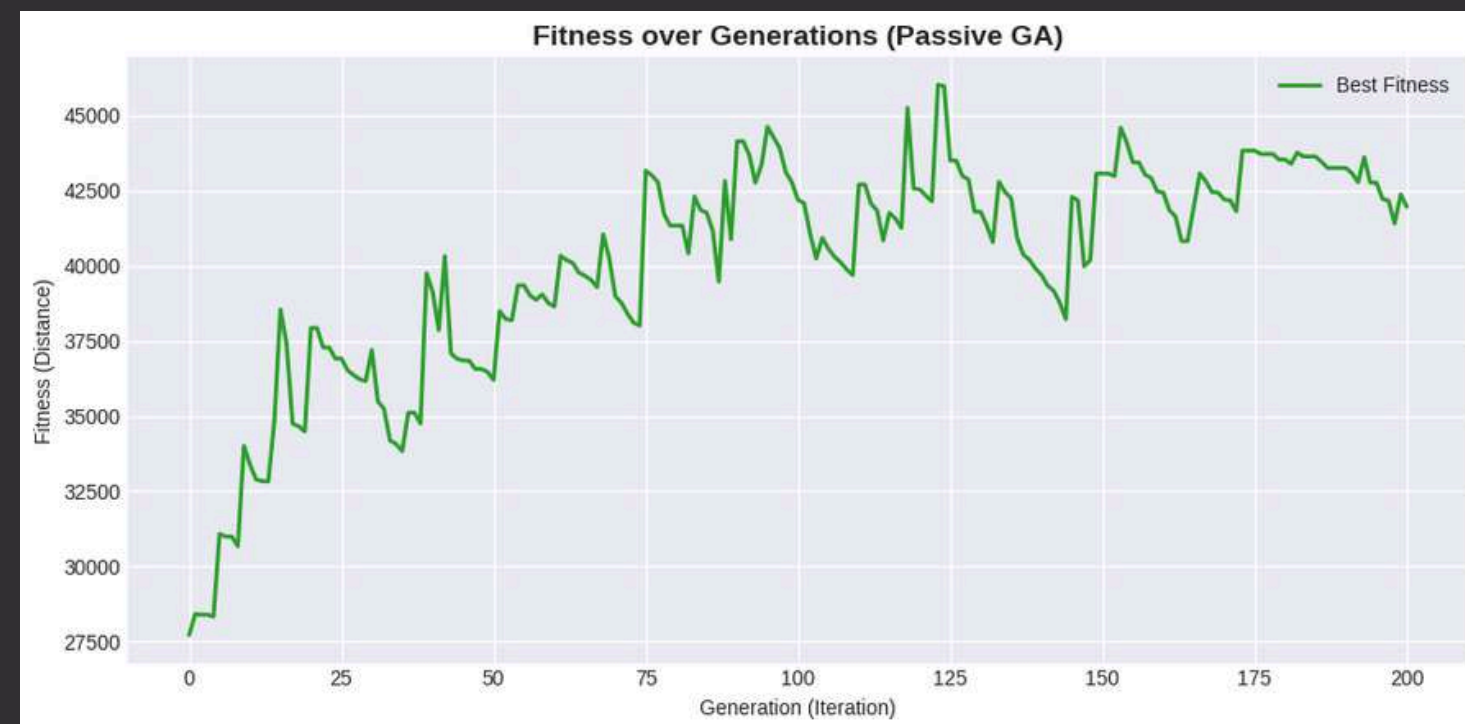
Phase 2: Deterministic Crowding and Multimodal Extraction



Extracted Niche	Routing Cost	Feasibility	Deviation vs. Niche 1
Niche 1 (Global Best)	27,325.97	Valid (Capacity Met)	—
Niche 2 (Alternative A)	27,552.99	Valid	+227.02 units
Niche 3 (Alternative B)	27,687.22	Valid	+361.25 units
Total Execution Time: 81.56 seconds			

GA PARAMETER SETTINGS

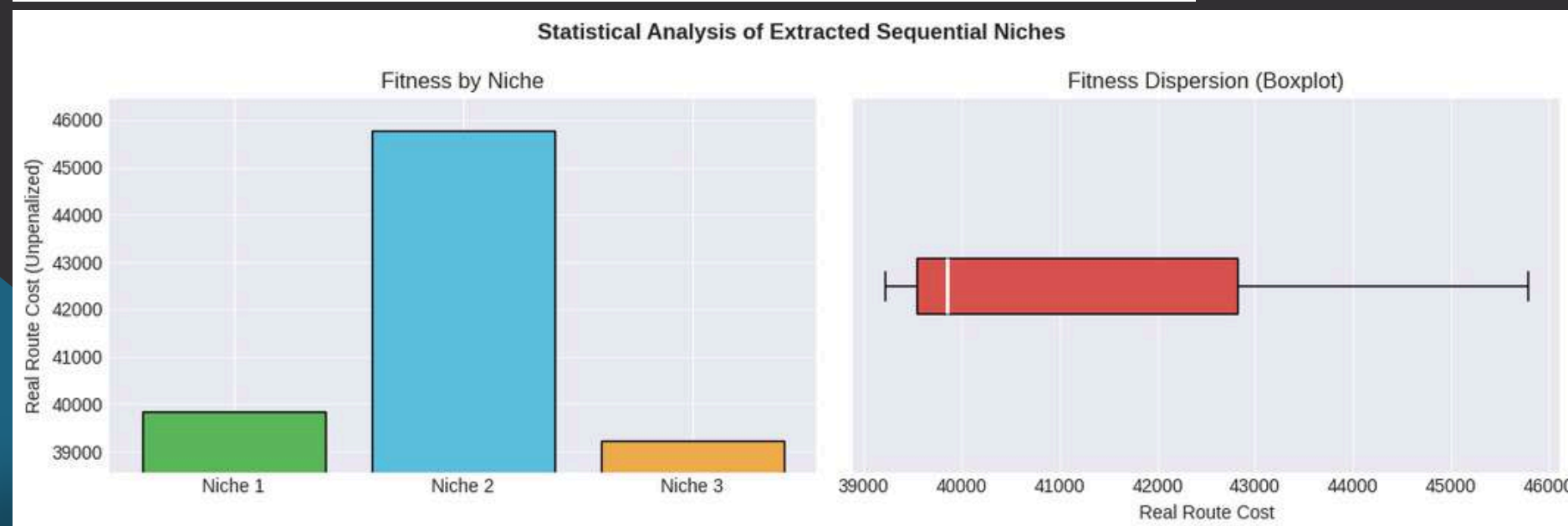
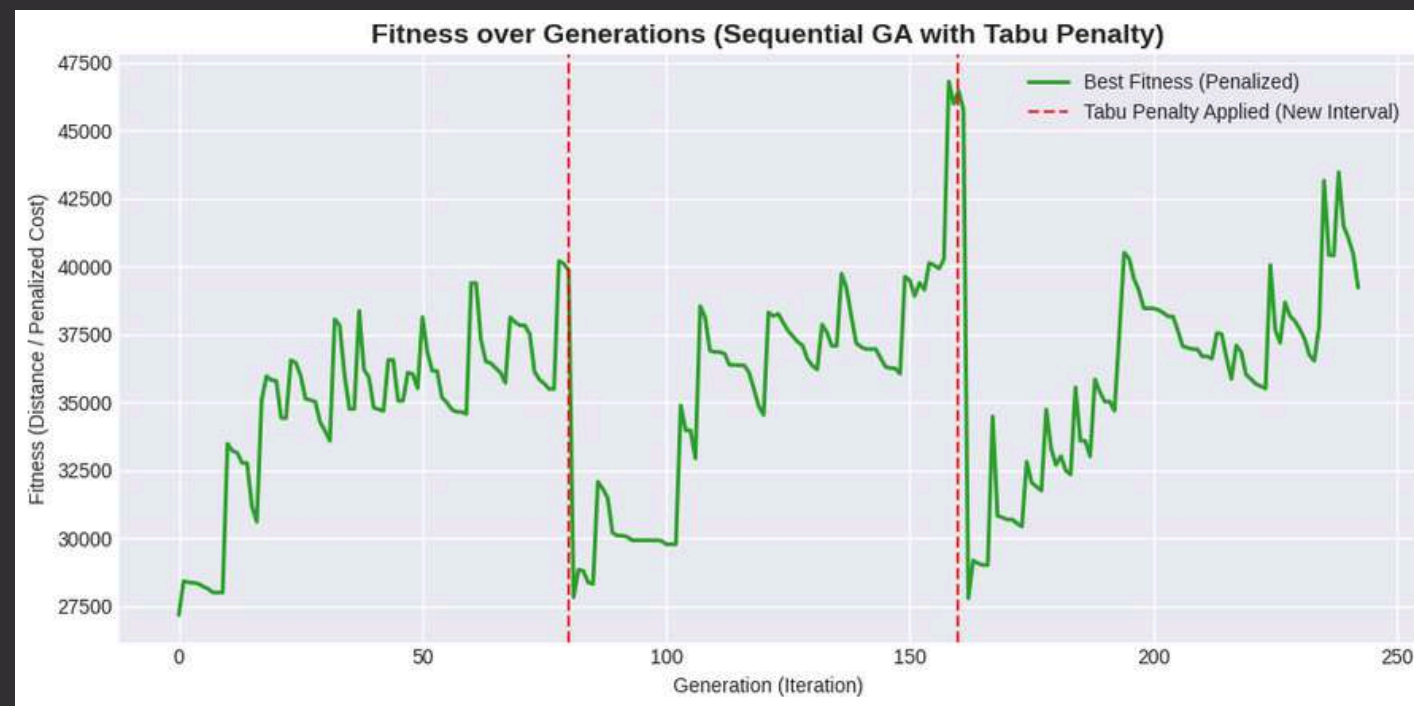
Phase 3: The Archive Method (Passive Niching & Soft Constraints)



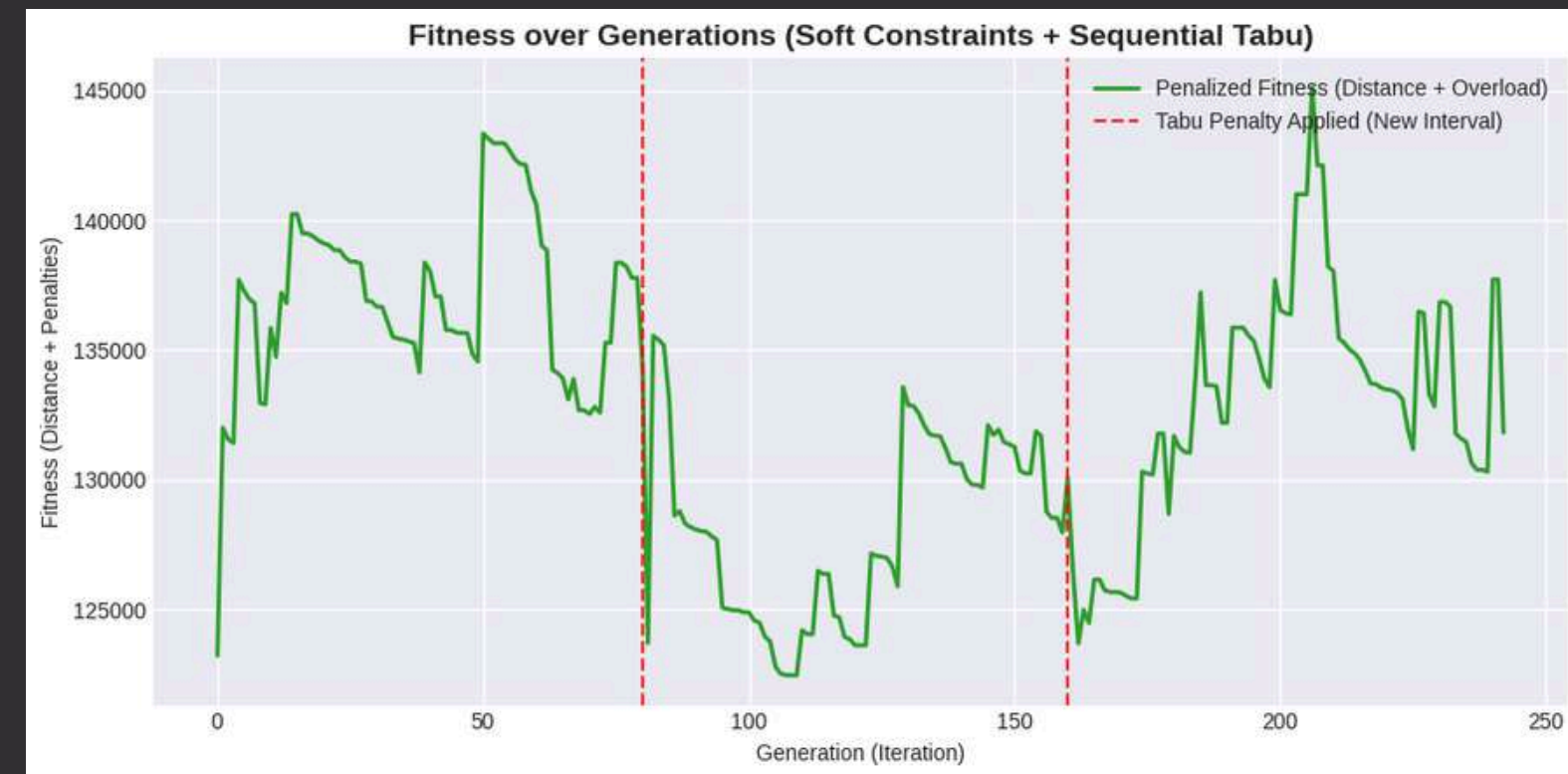
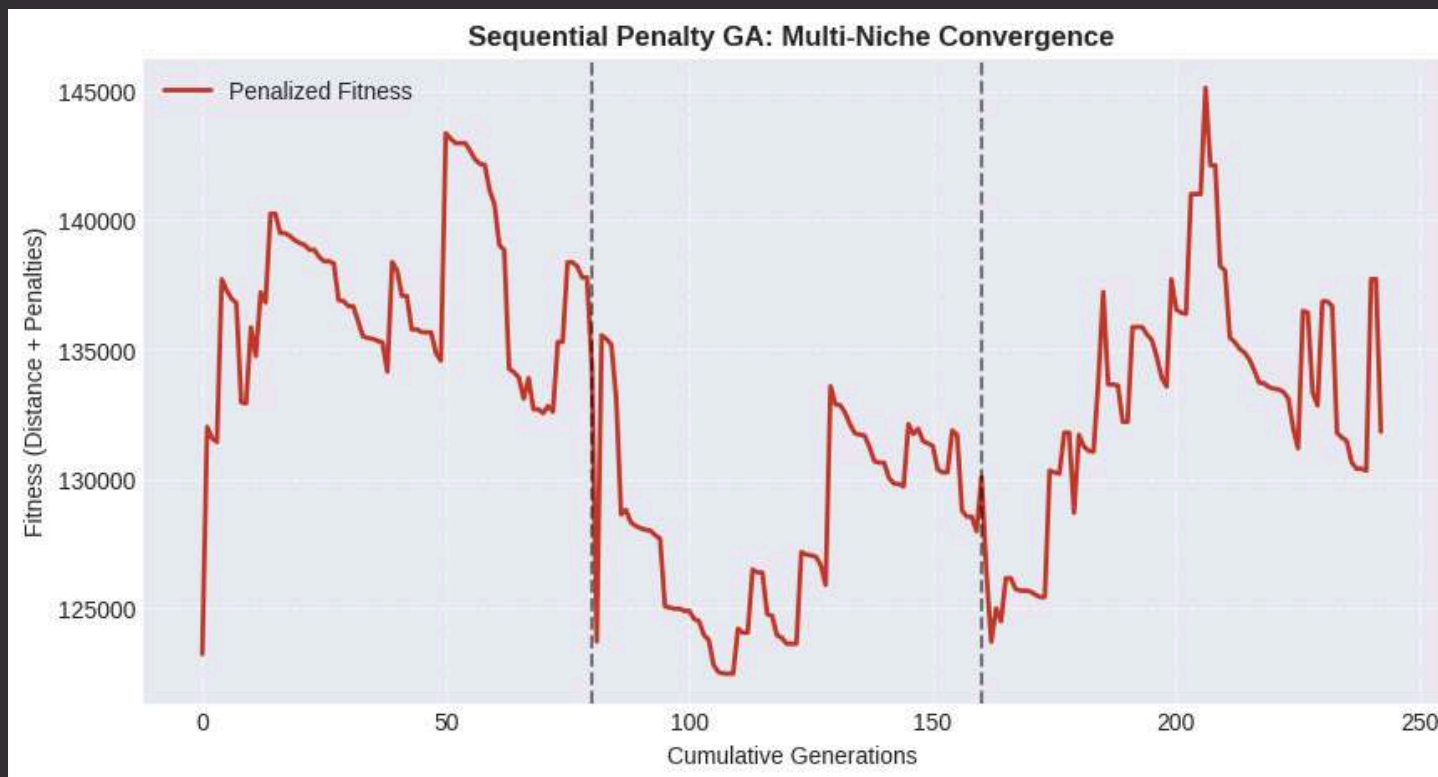
Extracted Niche	Routing Cost	Feasibility	Deviation vs. Niche 1
Niche 1 (Archive Best)	27,628.24	Valid (Healed)	—
Niche 2 (Alternative A)	27,679.52	Valid (Healed)	+51.28 units
Niche 3 (Alternative B)	27,764.12	Valid (Healed)	+135.88 units

GA PARAMETER SETTINGS

Phase 4: The Sequential Method (Adaptive Penalty & Tabu Search)



GA PARAMETER SETTINGS

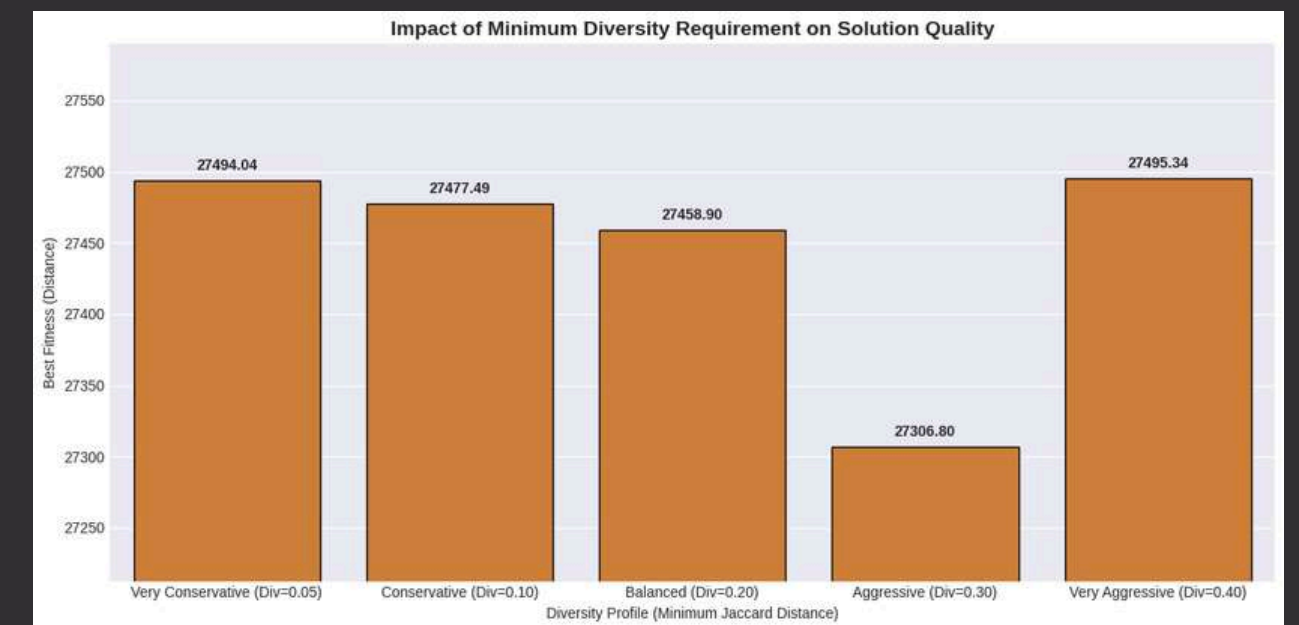
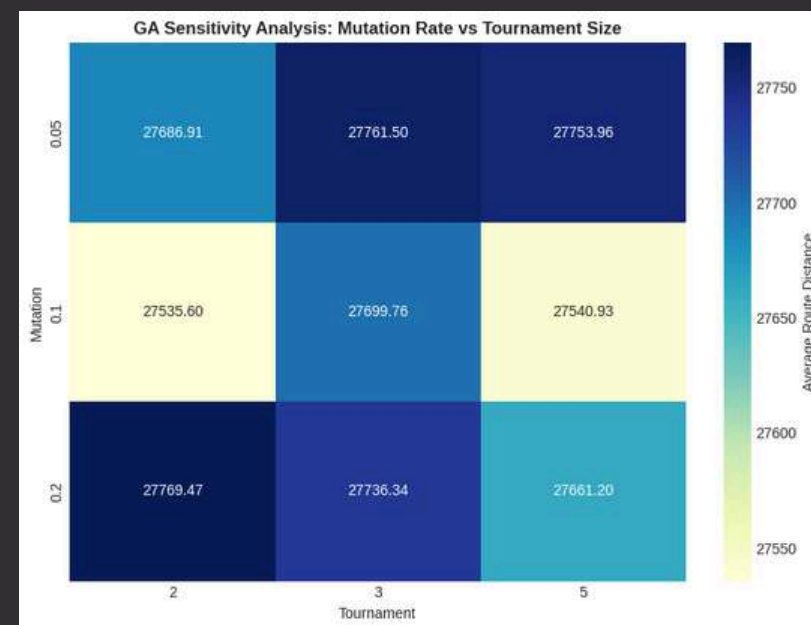


Pairwise Comparison	Jaccard Distance	Shared Structure	Status
Chaos Niche 1 vs. Niche 2	96.65%	~3.35%	Highly Diverse
Chaos Niche 1 vs. Niche 3	97.14%	~2.86%	Highly Diverse
Chaos Niche 2 vs. Niche 3	95.65%	~4.35%	Highly Diverse

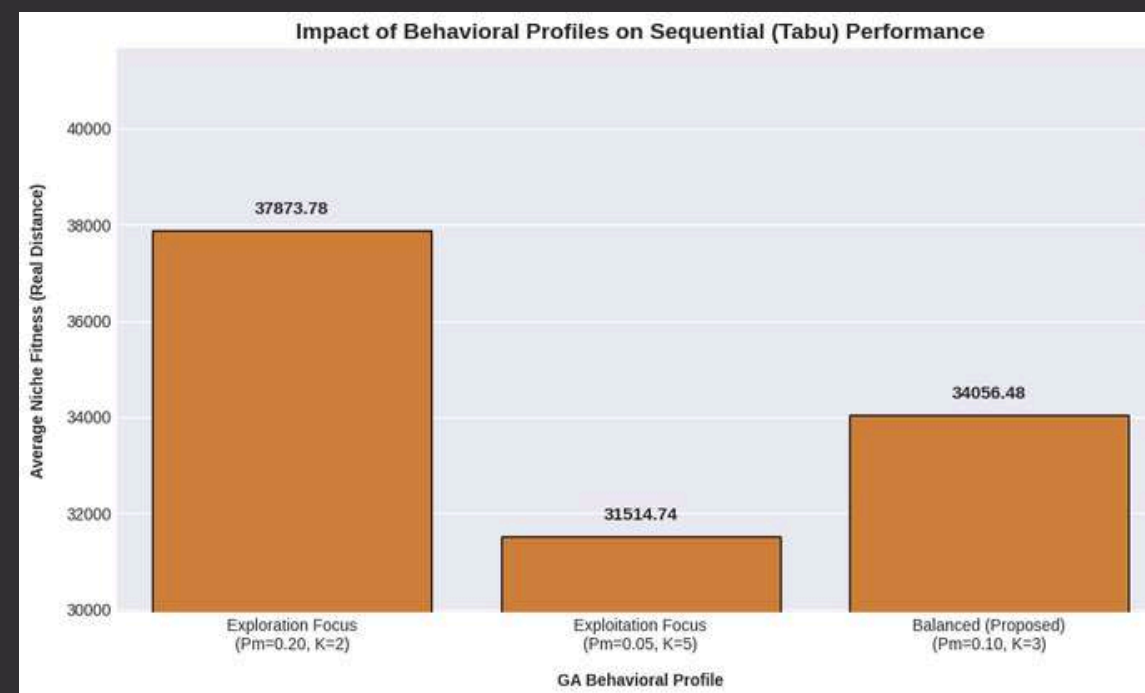
GA PARAMETER SETTINGS

Phase 5: Hyperparameter Tuning

Test A: Core Engine & Diversity Tuning (Feasibility Crowding)



Test B: Adaptive Tuning for Active Erasure (Sequential Tabu)



GA PARAMETER SETTINGS



Phase 6: Statistical Significance Testing

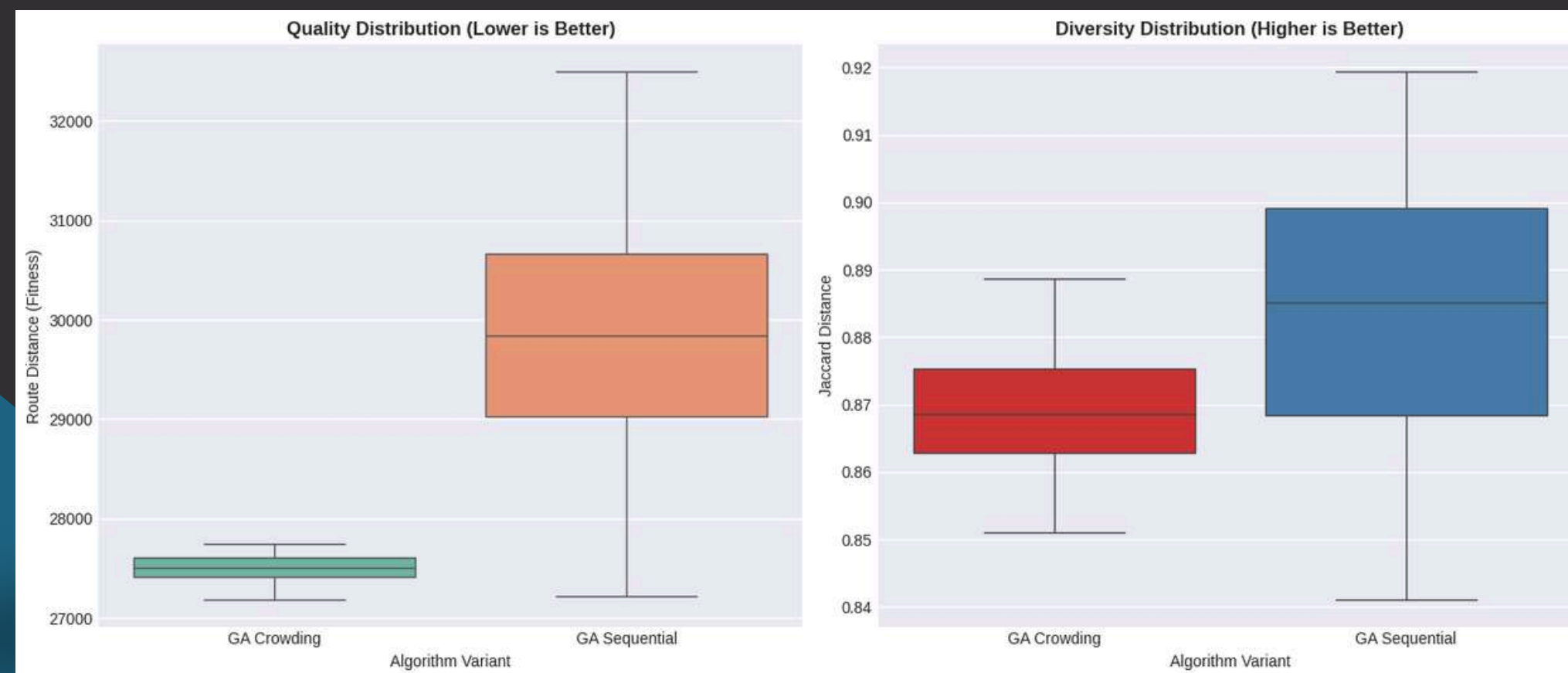
Algorithm	Mean Fitness ($\pm$ Std Dev)	Mean Jaccard	Execution Time
Feasibility Crowding	27,501.54 $\pm$ 136.59	86.92% $\pm$ 0.97%	173.38s / run
Sequential Tabu	29,909.69 $\pm$ 1312.55	88.38% $\pm$ 2.05%	45.66s / run

Aggregate results of the 35-Run final multimodal stress test

GA PARAMETER SETTINGS



Final Verdict: The Feasibility Crowding GA is the definitive champion for logistical application. By forcing competition strictly between structurally similar individuals, it consistently discovers highprecision routes that other methods overlook, maintaining an incredibly tight standard deviation (± 136). Conversely, while Sequential Tabu is computationally faster and guarantees higher maximum diversity, the conflict between strict Tabu restrictions and capacity-constrained geometries forces the algorithm into massive detours, severely inflating the operational cost and variance of secondary alternatives.



EXPERIMENTAL SETUP



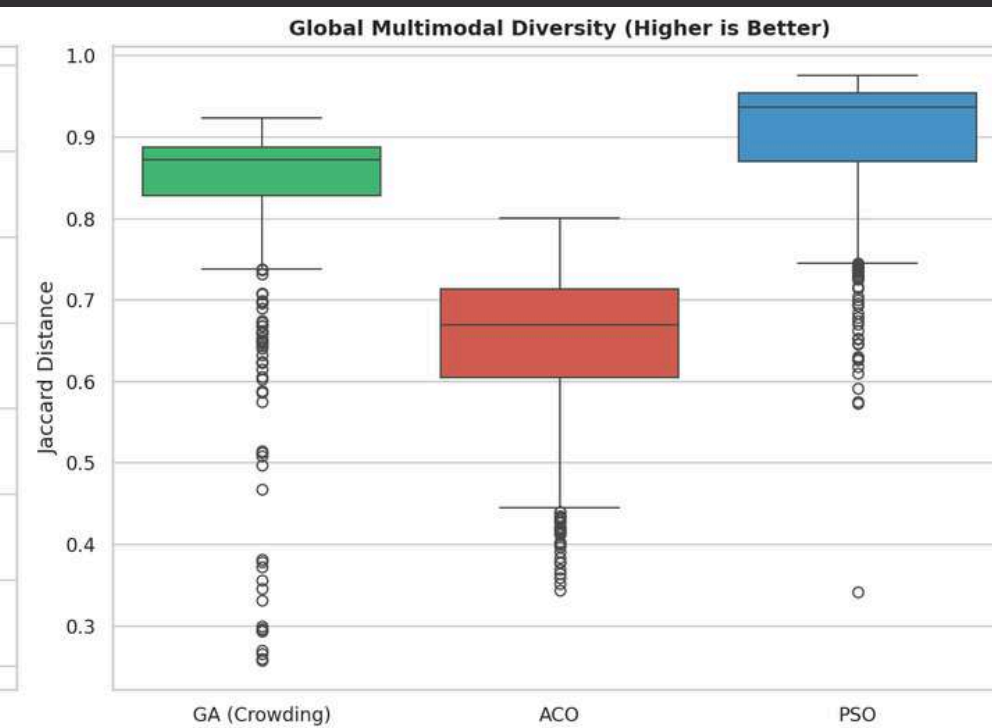
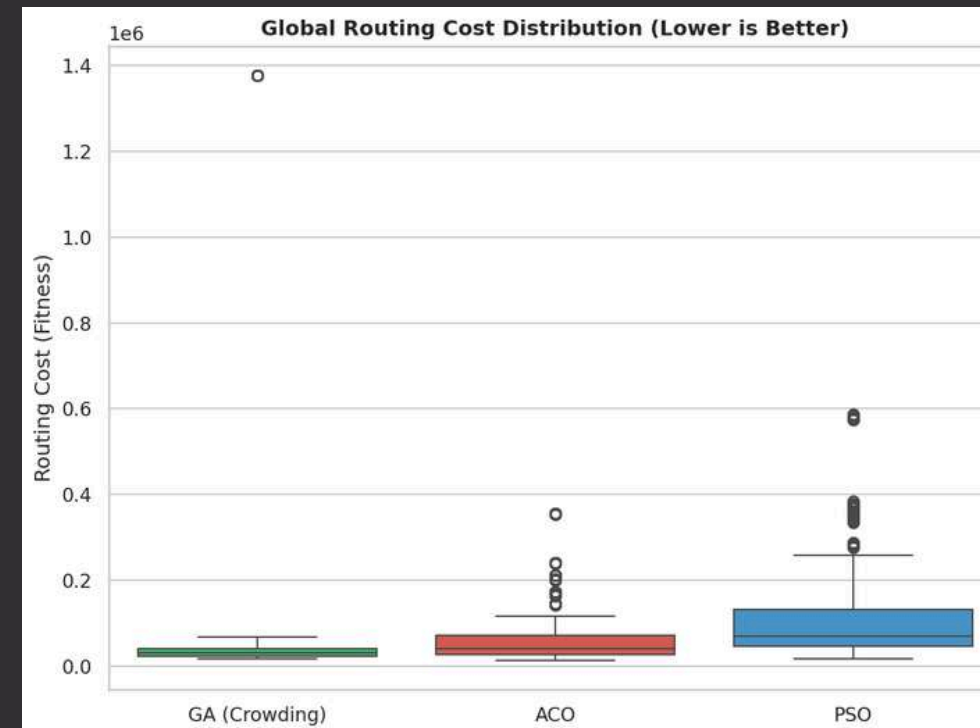
- Every instance was executed 35 independent times
- Having tuned and selected the optimal internal architecture for each metaheuristic, this final section evaluates the three undisputed champions against one another across the entire multiproblem dataset: the Complete Clustering PSO, the Feasibility Archive ACO, and the Feasibility Crowding GA. This global comparison evaluates the algorithms across the three fundamental pillars of multimodal optimization: Routing Quality (Fitness and Stability), Structural Diversity (Jaccard Distance), and Computational Efficiency (Runtime).

Algorithm	Avg. Routing Cost	Cost Std. Deviation	Avg. Diversity	Avg. Time (s)
ACO (Feas. Archive)	65,816.90	66,874.09	65.37%	323.80s
PSO (Complete Clustering)	112,095.67	112,399.83	90.61%	101.19s
GA (Crowding)	136,598.74	358,849.18	82.86%	16.03s

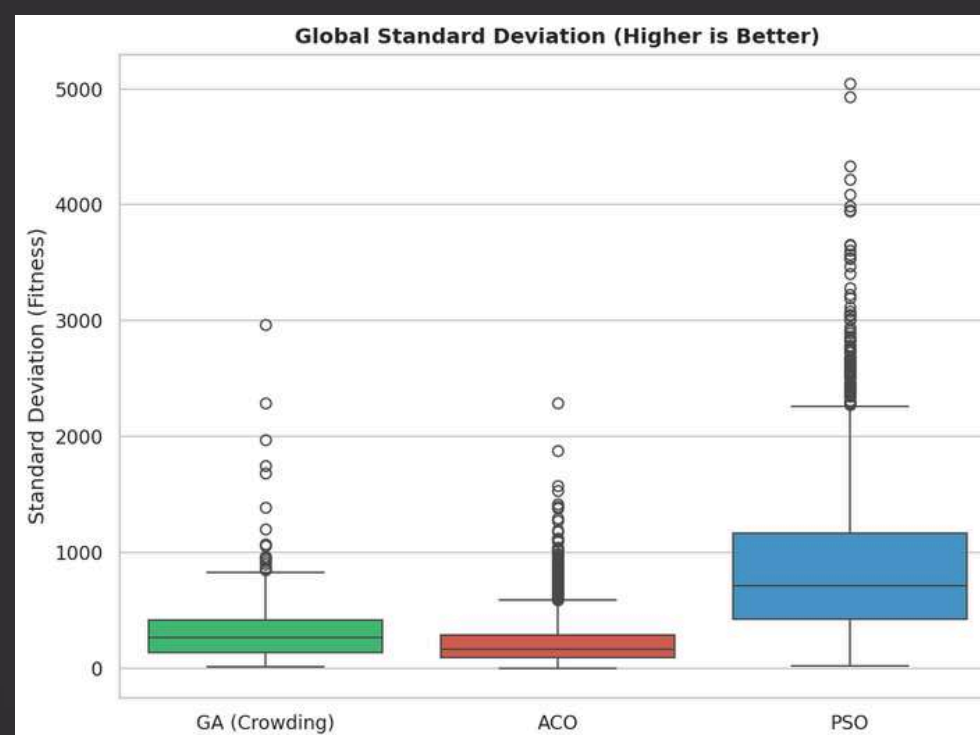
RESULTS



Routing Quality and
Stability Analysis



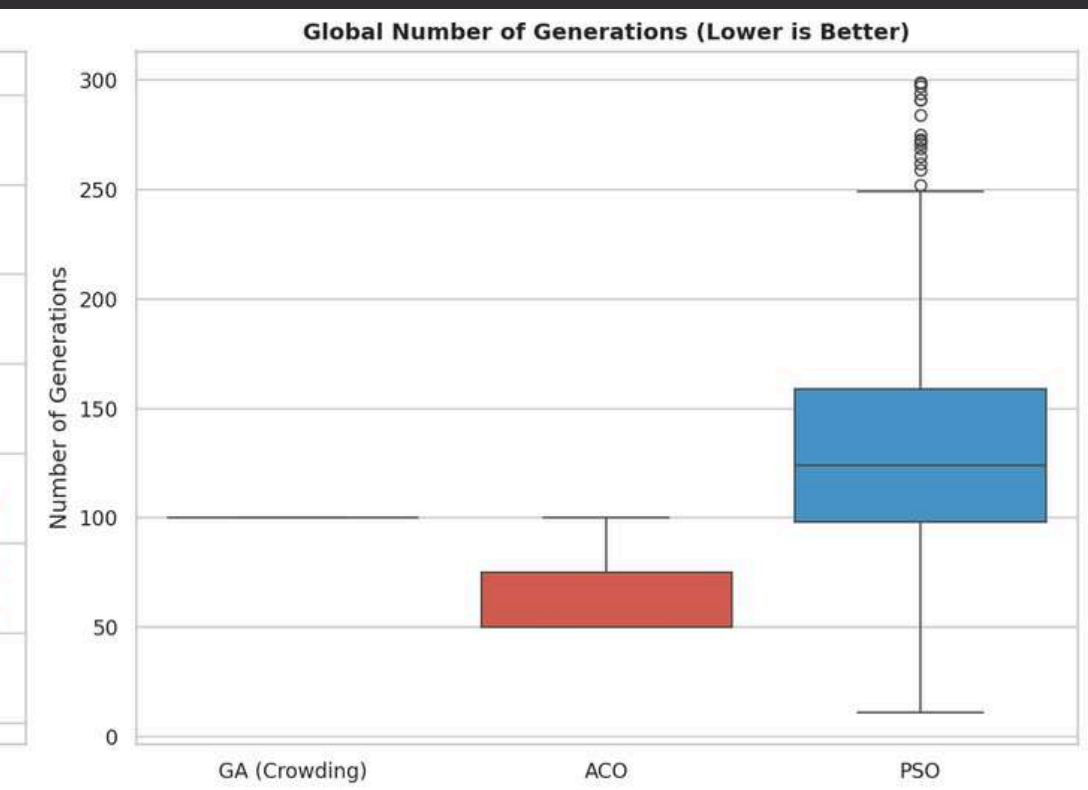
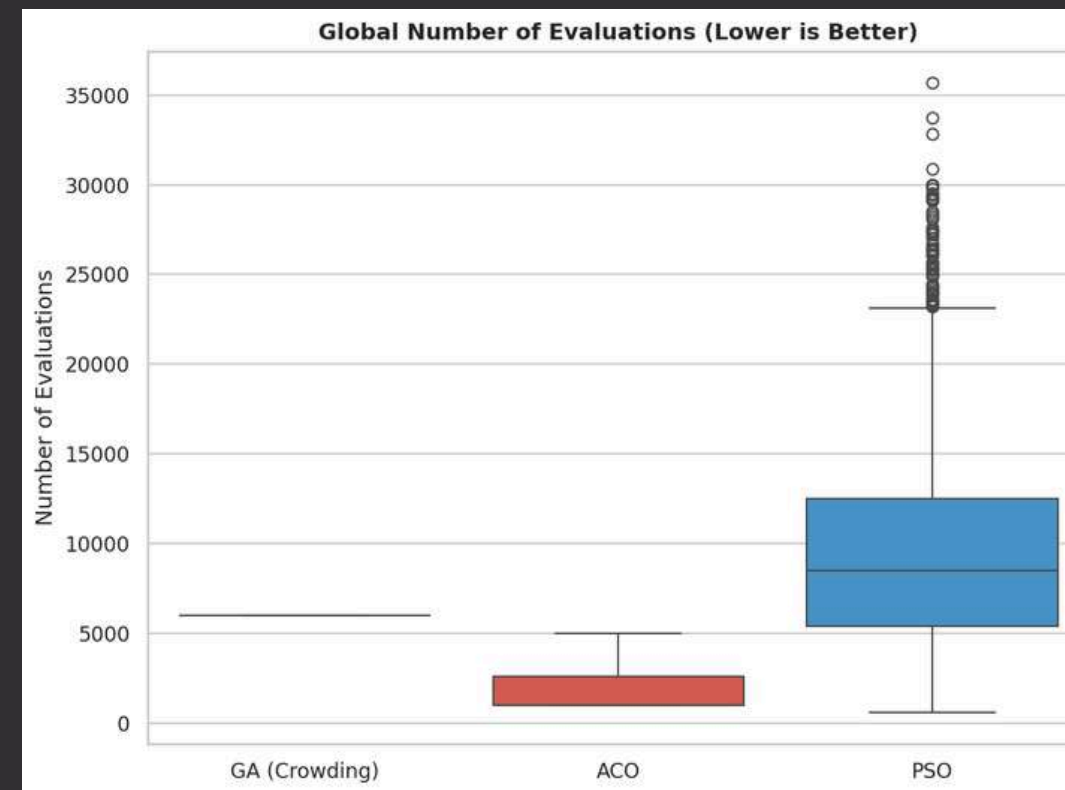
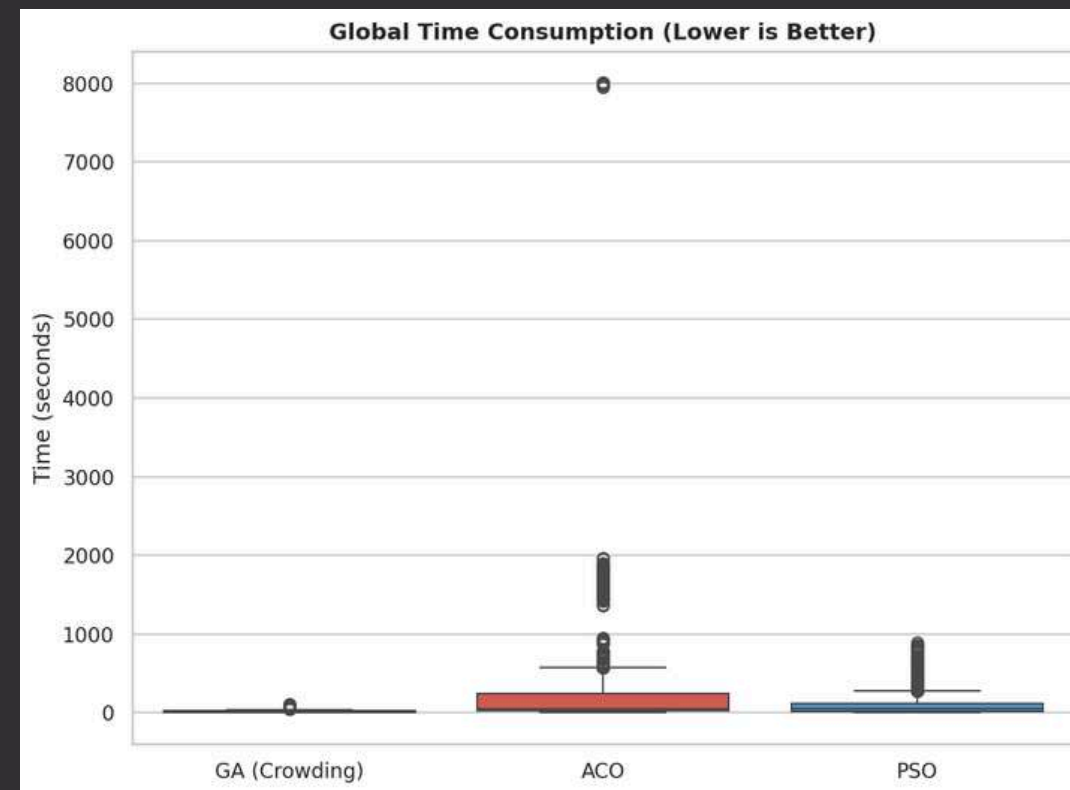
Multimodal Diversity
Analysis



RESULTS



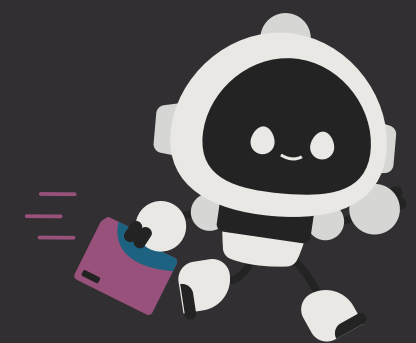
Computational Efficiency & Effort



CONCLUSIONS



The empirical data exposes a classic optimization trade-off. For mission-critical logistical operations where cost minimization and reliability are paramount, the Archive ACO is the unequivocally superior metaheuristic. However, if the operational priority shifts to requiring extreme structural contingencies (e.g., severe disaster routing) while maintaining moderate compute times, the Complete Clustering PSO offers the best architectural balance. Pure Genetic Algorithms, lacking native spatial awareness, are unsuited for large-scale CVRPs without heavy, custom hybridization





SOMEONE
WANTS TO ASK





THANK
YOU

